A Final Project Report On

Learniverse - Personalized learning through AI generated Tests

Submitted in the Partial Fulfillment of the

Requirements for the Degree of Bachelor of Computer Engineering/

Bachelor of Software Engineering Awarded by Pokhara University

Submitted By:

Biswash Dhungana [20180058]

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Supervisor:

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School of Engineering

Faculty of Science and Technology

POKHARA UNIVERSITY

August, 2024

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School of Engineering

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STUDENT'S DECLARATION

We hereby declare that this project work entitled **Learniverse** is based on our original work. All concepts, data, code, and any other work from external sources have been properly cited and referenced in accordance with the guidelines provided by School of Engineering, Pokhara University

We owe all the liabilities relating to the authenticity and originality of this project work and project report.

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SUPERVISOR'S RECOMMENDATION

This is to certify that this project report entitled **Learniverse** prepared and submitted by the below listed team of students in partial fulfillment of the requirements of the degree of Bachelor of Computer Engineering / Bachelor of Software Engineering awarded by Pokhara University, has been prepared and completed under my supervision.

I hereby recommend the same for acceptance by the School of Engineering, Pokhara University.

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EXTERNAL EXAMINER'S RECOMMENDATION

The undersigned certified that they have evaluated this project report entitled **Learniverse** submitted by **Biswash Dhunagana, Kripa Sarkee Nepali, Puja Sunuwar, Sristhi Lamichhane .**and their oral presentation for partial fulfillment of the degree of Bachelor of Computer Engineering/ Bachelor of Software Engineering and recommended to the School of Engineering, Pokhara University for acceptance of this project work/report.

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LETTER OF APPROVAL

This project report **Learniverse** submitted by **Biswash Dhunagana, Kripa Sarkee Nepali,Puja Sunuwar, Sristhi Lamichhane** for partial fulfillment of the degree of Bachelor of Computer Engineering/ Bachelor of Software Engineering has been accepted by the School of Engineering, Pokhara University upon the recommendations of Supervisor and with the approval by the following examiner.

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CERTIFICATE

The undersigned certify that they have read and recommended to the Department of Computer and Software Engineering for acceptance, a proposal report entitled "***Learniverse***" submitted by Biswash Dhungana, Puja Sunuwar, Kripa Sarki Nepali, Sristhi Lamichhane in partial fulfillment of the requirement for the bachelor's degree in computer engineering/software engineering.

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We extend our heartfelt gratitude to the academics and staff who provided insightful commentary and constructive criticism throughout our study and production phases. Their expertise and dedication were instrumental in shaping an LMS that meets our project's requirements. Special thanks to our supervisor, Dr. Deepanjal Shrestha, for his invaluable resources and insights, which greatly influenced the development of our application. We also appreciate the various platforms and tools that played a pivotal role in creating this educational platform, including learning management systems, content delivery networks, development environments, libraries, and frameworks. Their contributions were crucial in building robust and scalable solutions.

ABSTRACT

Learniverse is a learning management system designed for students preparing for the Nepal Engineering Council license exam in software engineering. The platform allows teachers to publish courses divided into chapters, each with detailed descriptions. Students enroll in these courses, study the material, and practice AI-generated multiple-choice questions (MCQs) specific to each chapter. This innovative approach ensures targeted practice and assessment, reinforcing the principle that "practice makes perfect." The methodology includes course publication, chapter descriptions, student enrollment, study, and immediate feedback on MCQ performance to guide improvement. To further enhance Learniverse, it is recommended to continuously improve AI algorithms, expand course offerings, integrate interactive learning features, seek user feedback, and form partnerships with educational institutions. These enhancements will ensure that Learniverse remains a vital resource for engineering license exam preparation in Nepal.

Keywords: Learniverse, personalized learning, AI-generated tests, learning management system, Nepal Engineering Council, software engineering, computer engineering, multiple-choice questions (MCQs), exam preparation, educational technology.

TABLE OF CONTENT

COPYRIGHT i

STUDENT'S DECLARATION ii

SUPERVISOR'S RECOMMENDATION iii

EXTERNAL EXAMINER'S RECOMMENDATION iv

LETTER OF APPROVAL v

ACKNOWLEDGEMENT vii

ABSTRACT viii

LIST OF FIGURES xii

CHAPTER 1 1

INTRODUCTION 1

1.1 Background 1

1.2 Problem Statement 1

1.3 Objective 1

1.4 Significance 2

1.5 Application 2

CHAPTER 2 3

LITERATURE REVIEW 3

2.1 Comparison Table 5

2.1.1 Research gap 6

CHAPTER 3 7

SYSTEM ANALYSIS AND DESIGN 7

3.1 Conceptual Framework 8

3.2 Workflow 9

3.3 Methodology 11

3.3.1 Research and Analysis 11
3.3.2 Design Phase 11
3.3.3 Development 11
3.3.4 AI Integration 11
3.3.5 Testing and Quality Assurance 13
3.3.6 Deployment and Rollout 13
3.3.7 Training and Support: 13
3.3.8 Evaluation and Iteration: 13
3.3.9 Continuous Improvement: 13
3.4 Generalized System Architecture 14
3.5 User Interface Model 14
3.5.1 End User Interface 14
3.6 Use Case Diagram 15
3.7 Entity-Relationship Diagram (ER) 17
3.8 Implementation diagram 19
3.9 Class Diagram 24
3.10 Deployment Diagram 26
CHAPTER 4 28
ANALYSIS AND RESULT 28
4.1 System Testing 28
4.2 Outcome 29
4.2 Evaluation 29
CHAPTER 5 31
CONCLUSION AND FURTHER DISCUSSION 31
5.1 Conclusion 31

5.2 Limitation and Future Recommendation 31
5.2.1 Limitation 31
5.2.2 Future Recommendation 32
REFERENCE 33
APPENDICES 34

LIST OF FIGURES

Figure 3.1 Conceptual diagram 8
Figure 3.2 Workflow diagram 10
Figure 3.3 Generalized System Architecture 14
Figure 3.4 End User Interface 15
Figure 3.5 Use case diagram 16
Figure 3.6 Entity relationship diagram 17
Figure 3.7 Implementation diagram 20
Figure 3.8 Class diagram 24
Figure 3.9 Deployment diagram 26

CHAPTER 1

INTRODUCTION

1.1 Background

The idea for Learniverse was born out of a desire to help students better prepare for the Nepal Engineering Council License Exam, especially those studying software and computer engineering. Many students struggle to find study materials and practice questions tailored to their needs, so we set out to create a solution. Learniverse uses AI to generate personalized multiple-choice questions based on the chapters of a course, giving students the chance to practice and solidify their understanding in a more targeted way. Our goal was to develop an engaging, adaptive learning platform that not only provides organized content but also supports students by continuously assessing their progress. Learniverse combines cutting-edge technology with educational needs to make exam preparation more effective and personalized.

1.2 Problem Statement

- In Nepal, students preparing for the Nepal Engineering Council (NEC) License Exam, particularly those in software and computer engineering, face significant challenges in finding tailored study materials and practice questions.
- Traditional resources often lack specificity and fail to align closely with the exam's evolving content and format, leaving students with inadequate preparation tools. This misalignment creates a gap in effective exam preparation, leading to increased stress and lower performance outcomes.

1.3 Objective

- To change the way students from Software and Computer Engineering stream prepare for the Nepal Engineering Council License exam.

- To create complimentary multiple choice questions based on individual course chapters so as to practice using content specific exercises in a manner that is closely aligned with their own learning style and exam outlines.

1.4 Significance

- Learniverse uses AI to generate personalized multiple-choice questions that have been curated for specific chapters of a course. This customisation ensures that students get practice on subjects being taught, and therefore makes their learning stronger.
- Lerniverse has made the exam preparation process simple & easy through its structured and adaptive learning approach.

1.5 Application

- Educational Institutions: Implementation in schools, colleges, and universities to enhance teaching and learning experiences.
- Online Learning Platforms: Integration into existing e-learning platforms to improve user engagement and outcomes.

CHAPTER 2

LITERATURE REVIEW

Education is changing greatly due to technology improvements, especially with regard to personalized learning and adaptive educational systems.This review of literature aims at examining the theoretical underpinnings, technological innovations and practical implementations of Artificial intelligence enabled learning management systems. A major focus in this study is on the chapter wise mcq generator for focused and personalized learning.

The literature surrounding Learning Management Systems (LMS) has seen significant attention across various domains, with a particular focus on their application in e-learning management, workplace training, and comparative studies in different geographic contexts.

De Oliveira (2016) provides an integrative review that highlights the critical role of LMS in enabling e-learning by facilitating communication and interaction between teachers and students within virtual environments. However, the study identifies gaps in the research, especially in terms of using IT for managing e-learning, and calls for further exploration of learning analytics as a future research direction . (Oliveria, May 2016)

In the workplace context, LMSs are recognized as essential platforms for the efficient delivery of educational content. Irfandi & Festiyed (n.d.) emphasize that the usability of an LMS is crucial for effective training, noting that a user-friendly system significantly enhances the learning experience and speeds up the training process. They also identify the challenges of acceptance and implementation in organizational settings, suggesting that addressing these challenges is key to successful LMS adoption . (Ifrandi, n.d.)

Rahman et al. (2023) explore the application of LMSs in the teaching and learning process, analyzing a substantial body of literature from recent years. Their findings underscore the widespread adoption of LMSs in higher education, particularly in response to the COVID-19 pandemic, which accelerated the need for robust online learning platforms . (Rahman, 2019)

Lastly, Turnbull et al. (2024) conduct a comparative analysis of research methodologies employed in LMS studies within Australia and China. They observe distinct differences in the approaches taken by researchers in these two countries, with Chinese studies favoring quantitative methodologies and confirmatory research, while Australian studies are more diverse in their methodological approaches. This review highlights the importance of understanding cultural and contextual differences in LMS research and offers insights for future collaborative projects in the Asia-Pacific region . (Turnbull, 2020)

Together, these studies present a comprehensive view of the current state of LMS research, while also identifying critical areas for future exploration, particularly in the management of e-learning and the adaptation of LMSs to different cultural contexts.

2.1 Comparison Table

Aspects	**Oliveira (LMS and E-learning Management)**	**Australia and China (LMS Research Methodology)**	**Various Studies (Use of LMS in Teaching and Learning)**	**Workplace (LMS in the Workplace)**	**Learniverse**
Features	Communication and interaction in virtual spaces	Comparison of research methodologies in LMS studies	Online content delivery, data mapping	Usability, acceptance, implementation, and barriers	Personalized learning, AI-driven quiz generation, content integration
Focus Area	IT-enabled e-learning, gaps in research on IT for e-learning management	Research methodologies (quantitative, qualitative, mixed methods)	Use of LMS in teaching and learning	Critical usability elements, implementation challenges in workplaces	AI-driven personalization, seamless user experience
Key Findings	Quantitative methods are common; research gaps in IT for e-learning	Chinese studies favor quantitative, confirmatory research	Significant post-2020 advancements, mapping relationships between studies	Usability is critical for effective training; notable increase post-COVID-19	Enhanced learner engagement and performance, adaptable learning styles
Citation	(de Oliveira, 2016)	(Turnbull et al., 2024)	(Rahman et al., 2023)	(Irfandi & Festiyed, n.d.)	-

2.1.1 Research gap

Despite the extensive research in digital learning platforms a significant research gap exists in the area of AI-driven personalized learning and automated quiz generation. Learniverse differentiates itself by specifically addressing this gap through several unique approaches. Unlike other studies, Learniverse focuses on offering AI-driven personalized learning experiences and automated quiz generation using a highly pre-trained model for generating multiple-choice questions. This specific technologies application is not commonly emphasized in exciting research. Furthermore, while broader studies often explore various aspects of research and methodologies, content delivery, and generally usability. Learniverse zeroes in on the seamless user experiences and personalized learning, setting it apart from the more generalized approaches of other platforms. The key findings of the Learniverse indicates a targeted effort to enhance learner engagement and performance through adaptive learning solutions, responding to the ongoing need for usability and effective implementation in digital learning environments. This focus underscores the necessary for more research in developing AI-driven, personalized learning tools that can substantially improve educational outcomes.

CHAPTER 3

SYSTEM ANALYSIS AND DESIGN

The system design and development approach for Learniverse was structured and comprehensive, incorporating both conceptual frameworks and practical tools such as workflow diagrams, use cases and class diagrams. Initially, an in-depth research and analysis phase was conducted to explore existing Learning management systems (LMS), AI technologies, and personalized learning and assessment by gathering user requirements through surveys, interviews and focus groups with educators, learners and stakeholders. In the design phase, the system architecture and technical specifications were defined, focusing on user interface designs, that emphasize usability, accessibility and personalized learning experiences. Wireframes and prototypes were developed to effectively visualize the platform's features and functionalities. During the development phase, core features like AI-powered quiz generation, adaptive learning experiences and integration with exciting online learning platforms were implemented. Agile development methodologies were employed to allow for iterative building, testing and refinement of the platform, ensuring that the system architecture was scalable, secure and interoperable.

3.1 Conceptual Framework

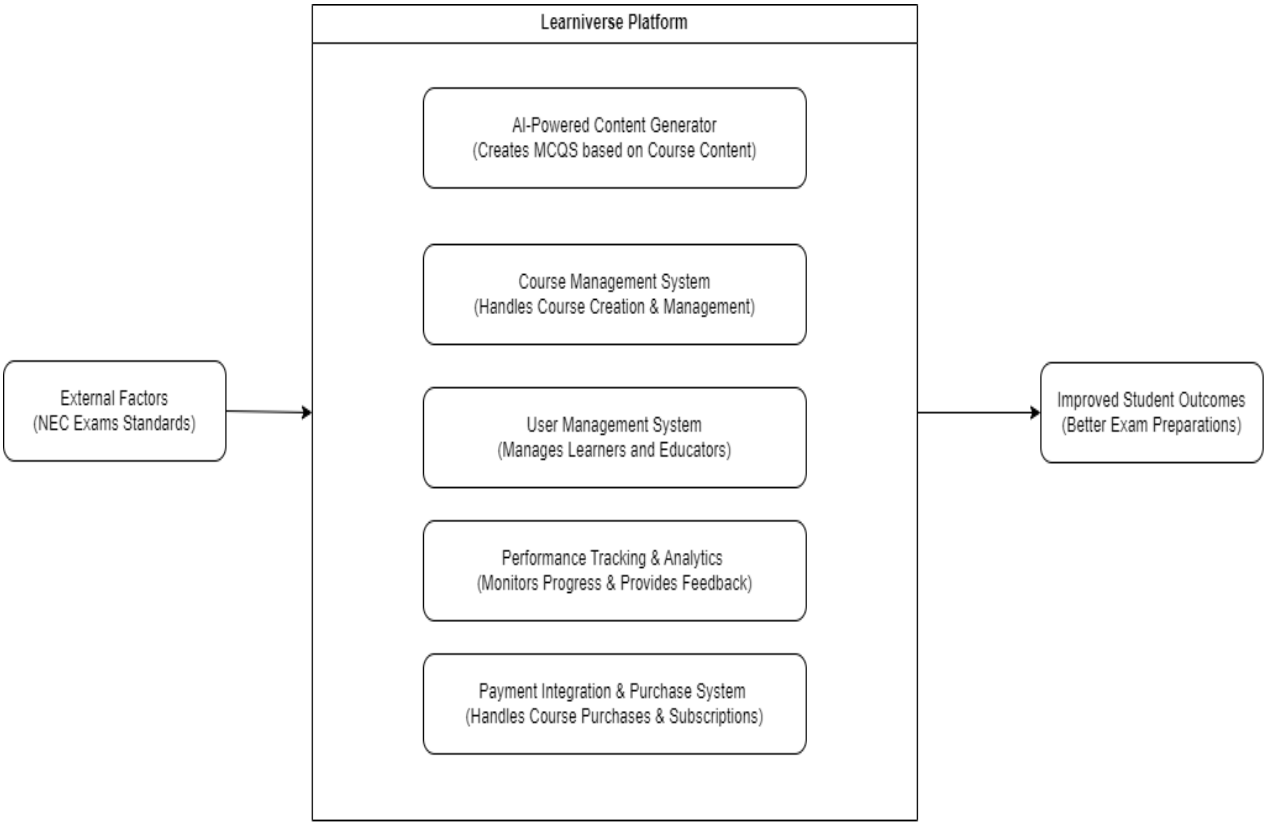

Figure 3.1 Conceptual diagram

In this work, we first acknowledge the significance of on-demand MCQ generation for a learning experience, course management system, user management system, performance tracking and analytics, and payment system integration as experienced by the research gap. The conceptual framework above illustrates the expected relationship between the variables needed for projects. It defines relevant objectives for the research process and maps out how they come together to draw coherent conclusions. The conceptual framework illustration above depicts cause-and-effect relationships.

- External factors: Represents external standards like NEC exam requirements that the system needs to align with.
- Learniverse Platform: This is the core of the framework, consisting of various interconnected systems:

- AI-Powered Content Generator: Creates personalized MCQs based on course content.
- Course Management System: Facilitates course creation and management.
- User Management System: Manages Learners and Educators within the platform.
- Performance Tracking and Analytics: Tracks user progress and provides analytical feedback to enhance learning.
- Payment Integration and Purchase System: Manages courses purchase and subscriptions.

- Improved Student Outcomes: The ultimate goal of the platform, resulting in better preparation for exams.

3.2 Workflow

Workflow represents the activities that was followed while building the Learniverse. From the starting this project with proper planning by defining the goals and objectives and dividing the task, then gathering the resources by collecting data for building the functional mockups and wireframes and by identifying the technologies implementing the app with the help of full stack Next.js framework which helps to build the client side and server side logics to make it work what we thought of and the highly pre-trained AI model is integrated in our app which helps in generation of MCQs, lastly we tested app by verifying functionality and performances and release the app.

Workflow for completion of our project Learniverse is as below:

- Plan: By defining goals and objectives and dividing tasks.
- Gather resources: By collecting data and identifying technologies.
- Design App: By Preparing wireframes and functional mocks ups.
- Implement App: By developing client and server side with Next.js.
- AI Integration: Integrated AI in routes for the MCQ generation.
- Testing: By verifying functionalities and Performance testing.
- Deploy App: With the help of Vercel.

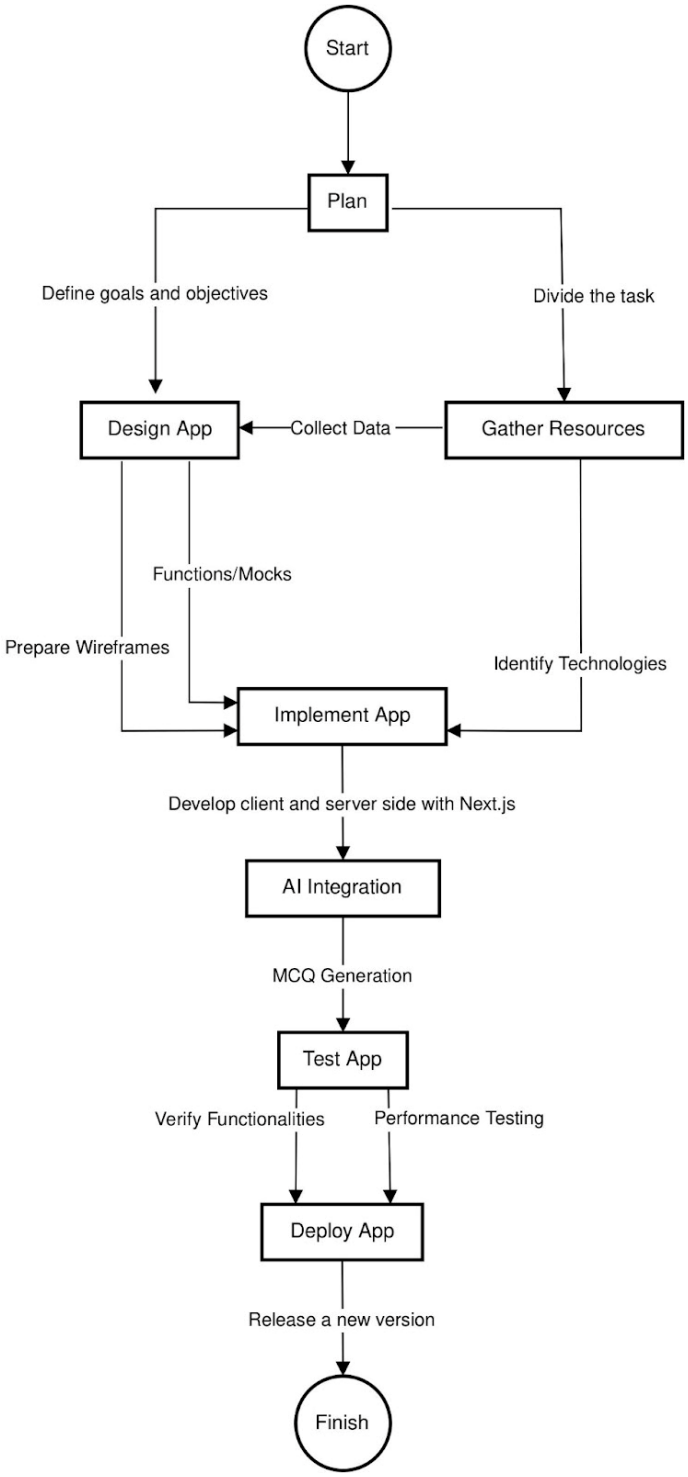

Figure 3.2 Workflow diagram

3.3 Methodology

3.3.1 Research and Analysis

- Conduct an in-depth analysis of existing LMS platforms, AI technologies, and personalized learning approaches.
- Identify key challenges and opportunities in personalized learning and assessment.
- Gather user requirements through surveys, interviews, and focus groups with educators, learners, and stakeholders.

3.3.2 Design Phase

- Define the system architecture and technical specifications for Learniverse.
- Design user interfaces that prioritize usability, accessibility, and personalized learning experiences.
- Develop wireframes and prototypes to visualize the platform's features and functionalities.

3.3.3 Development

- Implement the core features of Learniverse, including AI-powered quiz generation, adaptive learning experience, and integration with online learning platforms.
- Utilize agile development methodologies to iteratively build, test, and refine the platform.
- Ensure scalability, security, and interoperability of the system architecture.

3.3.4 AI Integration

Gemini's MCQs generation process is a complex interplay of natural language processing, knowledge retrieval, and question formulation techniques. Here's a simplified breakdown of the workflow:

- Prompt Understanding:
    - Input: A topic or subject matter is provided as a prompt.

    - Processing: Gemini's language model analyzes the prompt to identify key concepts, entities, and relationships.
- Knowledge Retrieval:
    - Search: Gemini searches its vast knowledge base (which includes text, code, and other forms of information) to find relevant information related to the prompt.
    - Selection: The model selects the most pertinent information based on relevance and quality.
- Question Formulation:
    - Question Types: Gemini can generate various types of questions, such as multiple-choice, fill-in-the-blank, or short answer.
    - Question Generation: Based on the retrieved information, the model crafts questions that test understanding, application, analysis, synthesis, or evaluation.
    - Distractor Generation: For multiple-choice questions, Gemini generates plausible incorrect answers (distractors) that may confuse or mislead test-takers.
- Question Refinement:
    - Clarity and Conciseness: The model ensures that the questions are clear, concise, and avoid ambiguity.
    - Difficulty Level: Gemini can adjust the difficulty level of the questions based on the target audience or specific requirements.
    - Relevance: The questions are double-checked to ensure they align with the original prompt and the retrieved information.
- Output:
    - MCQs: Gemini generates a set of MCQs, complete with questions and answer choices.
    - Feedback: The model can provide additional feedback or explanations for each question, if desired.

3.3.5 Testing and Quality Assurance

- Conduct comprehensive testing to identify and address bugs, errors, and usability issues.
- Perform user acceptance testing (UAT) with educators and learners to validate the effectiveness and usability of the platform.
- Ensure compliance with industry standards, data privacy regulations, and accessibility guidelines.

3.3.6 Deployment and Rollout

- Prepare deployment packages and documentation for installation and configuration of Learniverse.

3.3.7 Training and Support:

- Provide training sessions and resources for educators, administrators, and learners to effectively use Learniverse.
- Continuously update documentation and training materials based on user feedback and platform updates.

3.3.8 Evaluation and Iteration:

- Evaluate the effectiveness of Learniverse in improving learner engagement, performance, and satisfaction.
- Use insights from evaluation metrics to iterate and enhance the platform's features, algorithms, and user experience.

3.3.9 Continuous Improvement:

- Establish mechanisms for ongoing monitoring, maintenance, and updates to Learniverse.
- Stay abreast of advancements in AI technologies, educational research, and industry trends to incorporate new capabilities and innovations into Learniverse.

3.4 Generalized System Architecture

This is the system architecture that gives the sketch of how client side rendering, server side rendering is done and how server is interacting to the backend services. Here in Learniverse for client side rendering Next js is used and for server side rendering Nextjs api routes is used which interacts with backend services like Authentication with third party O-Auth library, AI Model Gemini for mcq generation, for data storage Mongodb Atlas , for video, image and file storage UploadThings and Stripe as a payment portal.

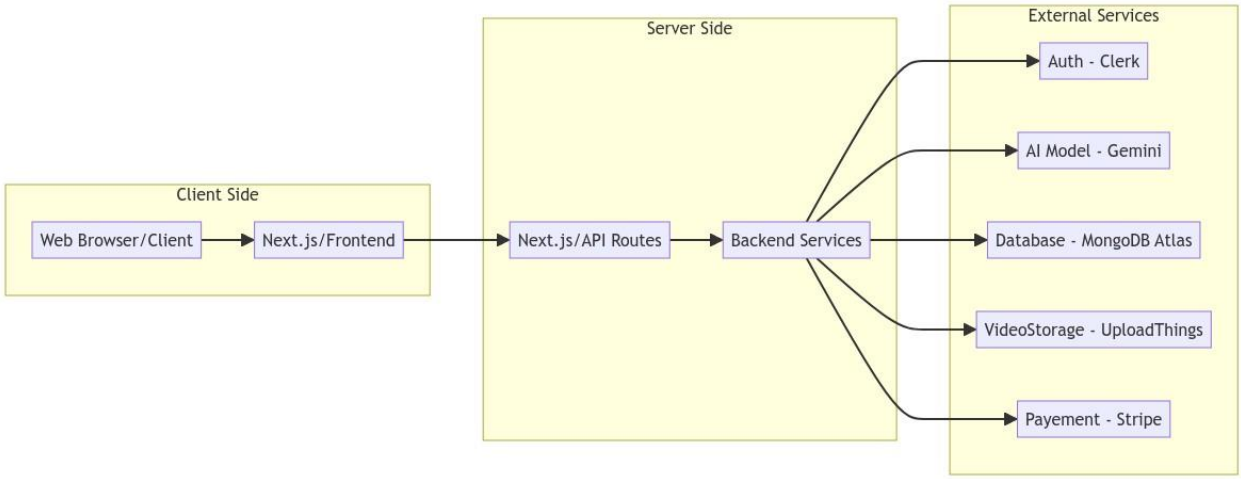

Figure 3.3 Generalized System Architecture

3.5 User Interface Model

The model which shows how the user interacts with the interface i.e Frontend components is call user interface model.

3.5.1 End User Interface

The Interface which is for the guest user is shown here as end user interface. After a successful login user will be redirected to the dashboard which consists of the courses. Inside course there are chapters. After buying the course then only user have access to the resources inside the chapter. Chapter have resources like videos, descriptions, attachment and MCQs button. After studying the provided content then user will generate mcq themselves, initially they have to

take beginner level mcq test, and only after scoring above 60% they will be able to take intermediate level of mcq otherwise have to retake the same level exams until scoring above that threshold value. And when user will score above 80% in intermediate level then they will be able to take the advance level mcq. Lastly the final score will be shown in the form of dialog box.

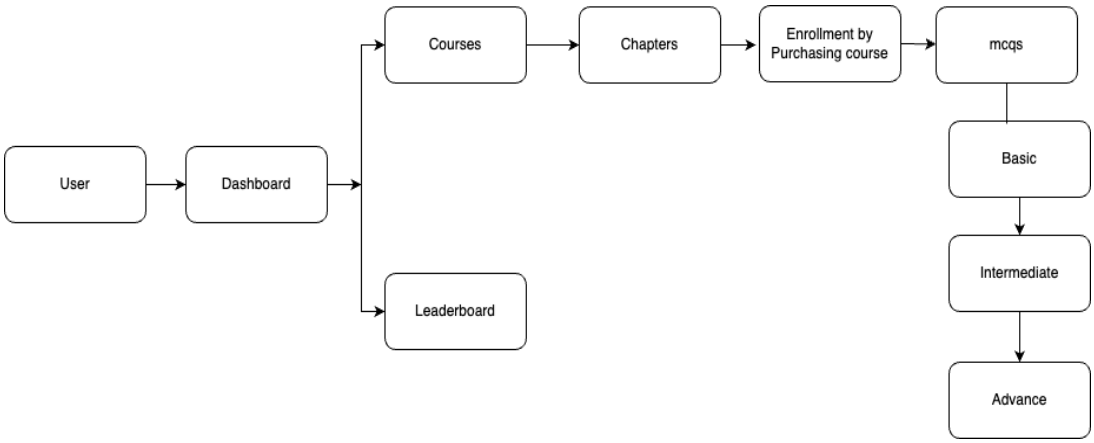

Figure 3.4 End User Interface

3.6 Use Case Diagram

The use case diagram is essential in the design and development process of Learniverse as it provides a clear and concise visualization of the interaction between users and the system. By mapping out various scenarios, such as how educators and learners will engage with features like AI-powered quiz generation and adaptive learning modules, the use case diagram helps in understanding the functional requirements of the system. This diagram also play a crucial role in identifying potential user roles, their goals and the different ways they can achieve those goals within the platform.

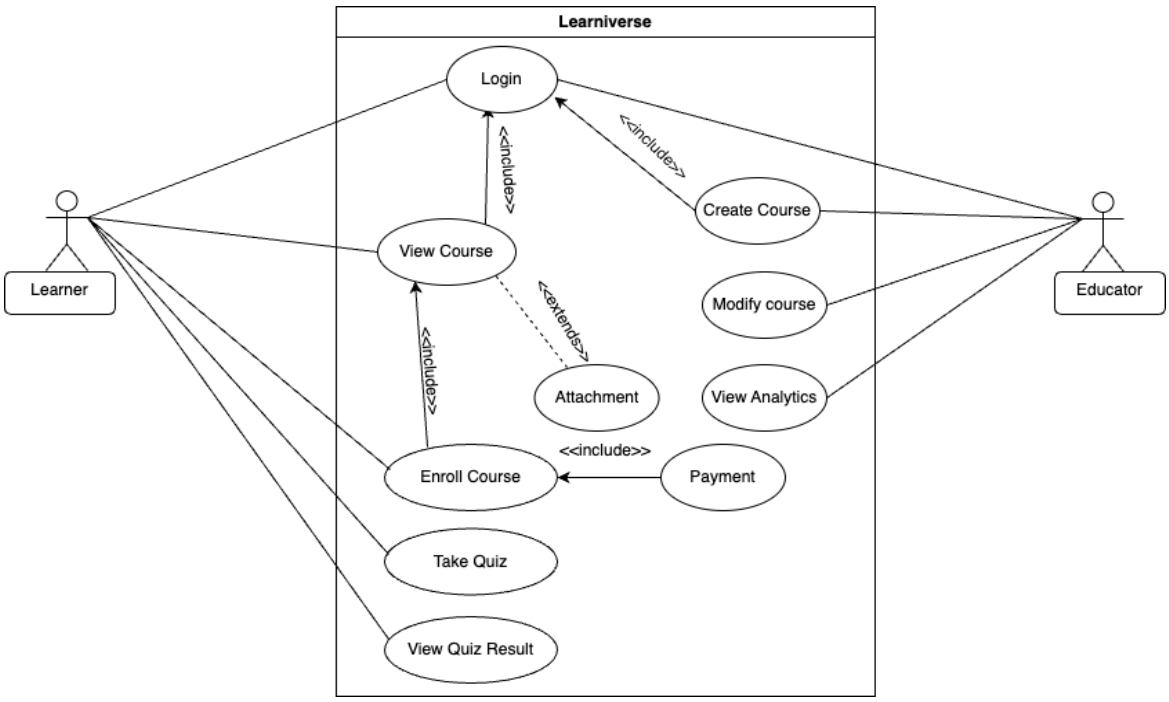

Figure 3.5 Use case diagram

- Educators:
  - Create Quizzes: Develop custom quizzes tailored to learning objectives, set parameters, and manage quiz settings.
  - Upload Content: Share educational materials like presentations, videos, and documents, organizing them into courses and modules.
  - Manage Courses: Define course objectives, schedule lessons and assessments, and monitor learner progress.
  - View Analytics: Access insights on learner performance and engagement to optimize teaching strategies.
- Learners:
  - Browse Content: Explore courses, lessons, and materials, searching for specific topics of interest.
  - Take Quizzes: Participate in quizzes assigned by educators, receiving immediate feedback on responses.

- ➢ Search Lessons: Quickly locate and access specific lessons or topics within courses for focused learning.

3.7 Entity-Relationship Diagram (ER)

ER diagrams are related to data structure diagrams (DSDs), which focus on the relationships of elements within entities instead of relationships between entities themselves. ER diagrams also are often used in conjunction with data flow diagrams (DFDs), which map out the flow of information for processes or systems.

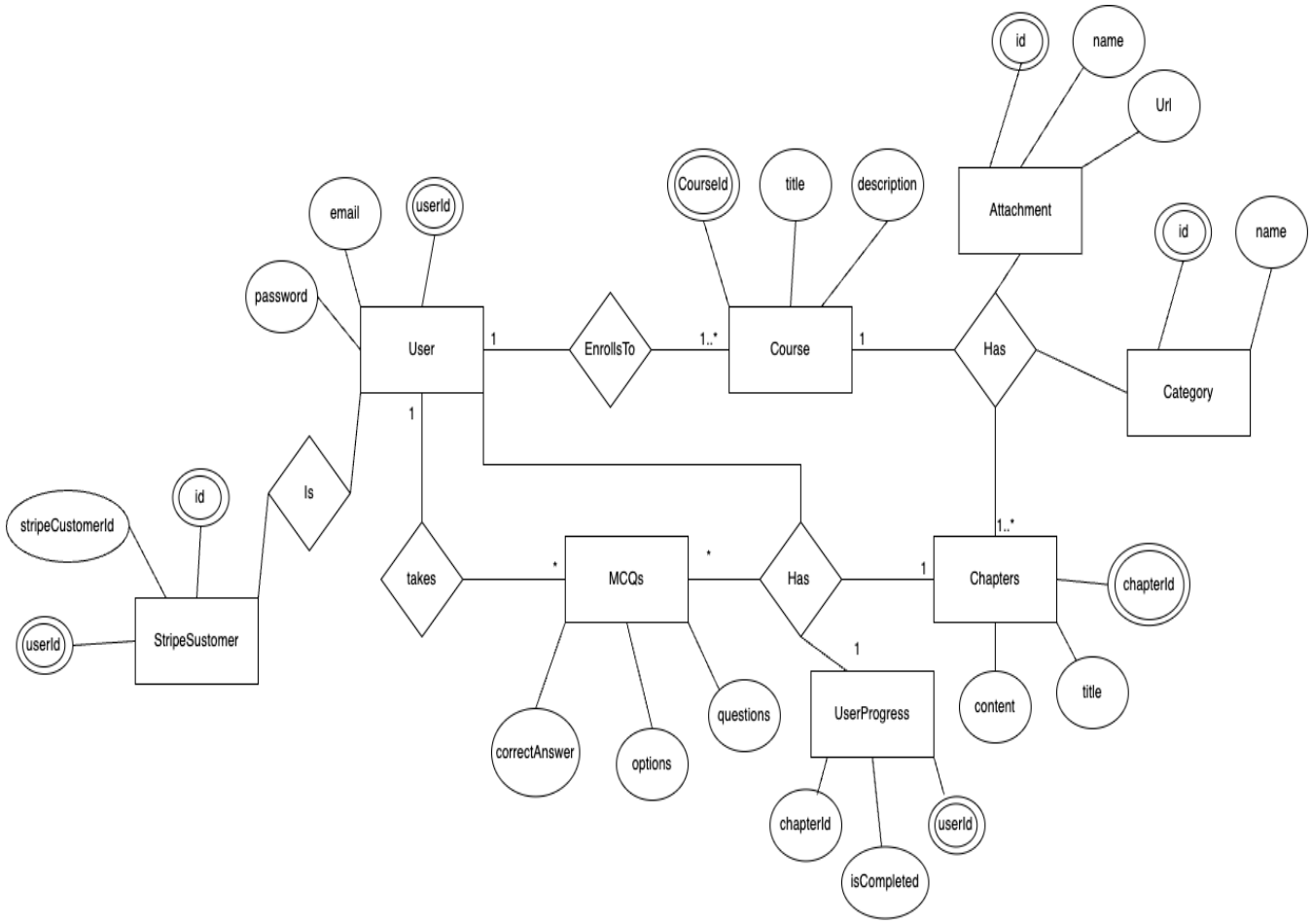

Figure 3.6 Entity relationship diagram

Entities:

- User
  - Attributes:
    - userId (Primary Key)
    - email
    - password
  - Represents users of the platform, including students and potentially teachers.
- Course
  - Attributes:
    - courseId (Primary Key)
    - title
    - description
    - isFree (Boolean indicating whether the course is free or paid)
  - Represents the courses available on the platform.
- Chapters
  - Attributes:
    - chapterId (Primary Key)
    - title
    - content
  - Represents the chapters within a course.
- MCQs
  - Attributes:
    - questions
    - options
    - correctAnswer
  - Represents multiple-choice questions related to a course or chapter.
- UserProgress
  - Attributes:

        - userId
        - chapterId
        - isCompleted (Tracks whether a user has completed a chapter)
    - Tracks user progress for each chapter.

Relationships:

- User EnrollsTo Course
    - A user enrolls in one or more courses, establishing a many-to-many relationship between the User and Course entities.
- Course Contains Chapters
    - A course contains one or more chapters, which establishes a one-to-many relationship between the Course and Chapters entities.
- User Takes MCQs
    - A user takes multiple-choice questions (MCQs) associated with a course. This is a many-to-many relationship between User and MCQs.
- MCQs Has Questions and Answers
    - MCQs have associated questions, options, and the correct answer, forming a one-to-many relationship within the MCQs entity.
- Chapters Has UserProgress
    - Each chapter tracks the progress of different users, indicating a one-to-many relationship between Chapters and UserProgress. One chapter can have multiple progress records associated with different users.

3.8 Implementation diagram

Implementation diagram is the diagram that shows the implementation details in different layers of our architecture like in case of Learniverse there are three layers' user interface layer, Business logic layer and Data access layer.

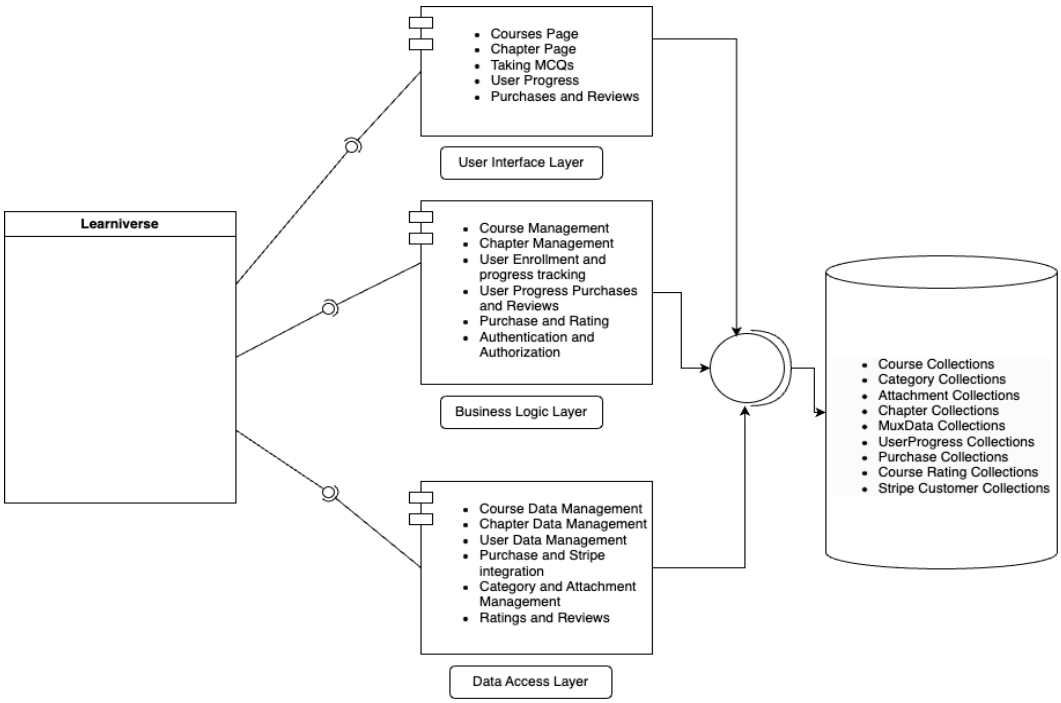

Figure 3.7 Implementation diagram

- User Interface (UI) Layer

  The User Interface layer is where users interact with the application. This layer doesn't directly deal with the Prisma models, but it would present data from the application and accept input from users. Here's how it interacts with the models:

  - Displaying Courses: Users would see a list of courses (Course), including titles, descriptions, prices, and ratings.
  - Viewing Chapters: Users can view the chapters (Chapter) of a course, including details like the title, description, and video content.
  - Taking Quizzes: Users interact with MCQs (MCQ) related to chapters.
  - User Progress: Progress indicators might be shown based on the UserProgress model, indicating which chapters have been completed.
  - Purchases and Reviews: Users can purchase courses (Purchase) and leave ratings and reviews (CourseRating).

- Business Logic Layer

  The Business Logic layer processes user input, applies business rules, and updates the data accordingly. This layer will contain the core application logic that uses these Prisma models:

  - Course Management: Handling the logic for creating, updating, and deleting courses (Course), as well as publishing or unpublishing them.
  - Chapter Management: Managing the addition and arrangement of chapters (Chapter), ensuring they are linked to the correct course and possibly handling video uploads and mux data (MuxData).
  - User Enrollment and Progress Tracking: Enrolling users in courses, tracking their progress through UserProgress, and updating progress as users complete chapters.
  - Purchases and Ratings: Handling the logic for course purchases (Purchase) and managing user ratings and reviews (CourseRatting).
  - Authentication and Authorization: Ensuring that only authorized users can modify data, such as course content, chapters, or reviews.

- Data Access Layer

  The Data Access layer involves interactions with the database, handled by Prisma in this case. It abstracts the database interactions and provides an interface for the business logic layer to query and manipulate data:

  - Course Data Management: CRUD operations on the Course model, including fetching, creating, updating, and deleting courses.
  - Chapter Data Management: CRUD operations on the Chapter model, including managing chapter order (position) and chapter-specific data like video URLs and descriptions.
  - User Data: Handling user-specific data such as UserProgress to track which chapters have been completed.

- Purchase and Stripe Integration: Managing Purchase records and integrating with the StripeCustomer model for payment processing.
- Category and Attachment Management: Managing Category associations and Attachment records linked to courses.
- Ratings and Reviews: CRUD operations on the CourseRatting model, allowing users to leave feedback on courses.

- Database Collections
  - Course Collection
    - Fields: id, userId, title, description, imageUrl, price, totalRating, isPublished, categoryId, createdAt, updatedAt
    - Relationships:
      - Category (One-to-Many): categoryId references Category collection.
      - Chapters, Attachments, Purchases, CourseRatings, MCQs (One-to-Many): Each references the Course collection.

  - Category Collection
    - Fields: id, name
    - Relationships:
      - Courses (One-to-Many): A category can have multiple courses.
  - Attachment Collection
    - Fields: id, name, url, courseId, createdAt, updatedAt
    - Relationships:
      - Course (Many-to-One): courseId references the Course collection.
  - Chapter Collection
    - Fields: id, title, description, videoUrl, position, isPublished, isFree, courseId, createdAt, updatedAt
    - Relationships:
      - Course (Many-to-One): courseId references the Course collection.

        - MuxData (One-to-One): chapterId references the MuxData collection.
        - UserProgress (One-to-Many): Each chapter can have multiple UserProgress documents.
    - MuxData Collection
        - Fields: id, assetId, playbackId, chapterId
        - Relationships:
            - Chapter (One-to-One): chapterId references the Chapter collection.
    - UserProgress Collection
        - Fields: id, userId, chapterId, isCompleted, createdAt, updatedAt
        - Relationships:
            - Chapter (Many-to-One): chapterId references the Chapter collection.
            - User (Many-to-One): userId references the User collection.
    - Purchase Collection
        - Fields: id, userId, courseId, createdAt, updatedAt
        - Relationships:
            - Course (Many-to-One): courseId references the Course collection.
            - User (Many-to-One): userId references the User collection.
    - CourseRating Collection
        - Fields: id, rating, review, userId, courseId, createdAt, updatedAt
        - Relationships:
            - Course (Many-to-One): courseId references the Course collection.
            - User (Many-to-One): userId references the User collection.
    - StripeCustomer Collection
        - Fields: id, userId, stripeCustomerId, createdAt, updatedAt
        - Relationships:
            - User (One-to-One): userId references the User collection.

3.9 Class Diagram

This class diagram illustrates the relationships between four main classes: User, Course, Attachment, and MCQ.

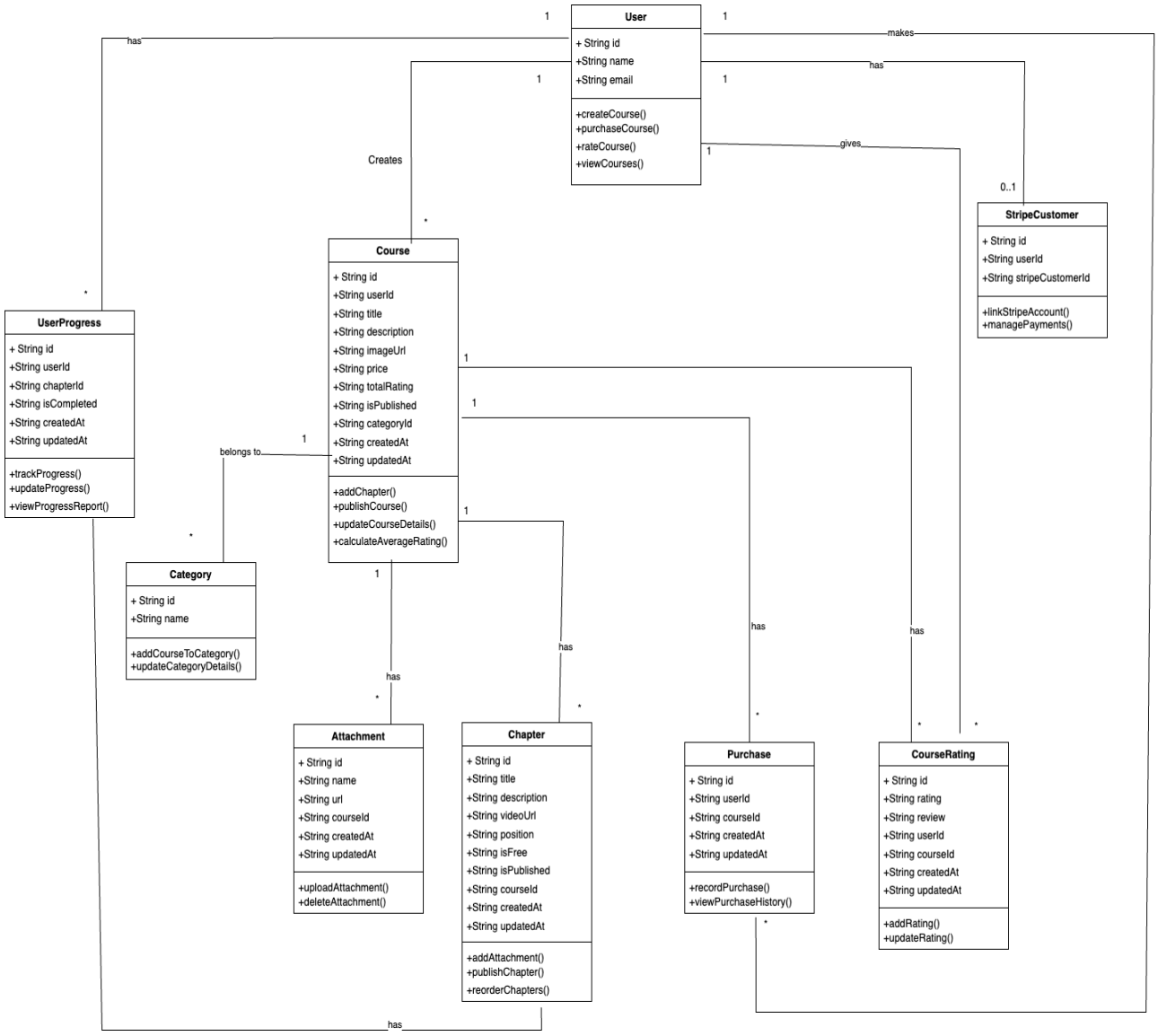

Figure 3.8 Class diagram

- User Class:
  - Attributes: id, name, email, password, isAuthenticated.

    - Relationship: A User can manage multiple Course instances (one-to-many relationship).
- Course Class:
    - Attributes: id, title, description.
    - Relationships:
        - A Course has multiple Attachment instances (one-to-many relationship).
        - A Course can include multiple MCQ instances (one-to-many relationship).
        - A Course belongs to a single User (many-to-one relationship).
- Attachment Class:
    - Attributes: id, fileName, fileType, filePath, courseId.
    - Relationship: An Attachment belongs to a single Course (many-to-one relationship).
- MCQ Class:
    - Attributes: id, question, choices, correctAnswer, courseId.
    - Relationship: An MCQ belongs to a single Course (many-to-one relationship).

3.10 Deployment Diagram

This deployment model outlines the flow of data and interactions between the client-side device, network, application server, and database server. The client-side device communicates with the application server through a T-1 network interface, which connects to the internal network. The application server processes user requests and interacts with the database server to retrieve or store data.

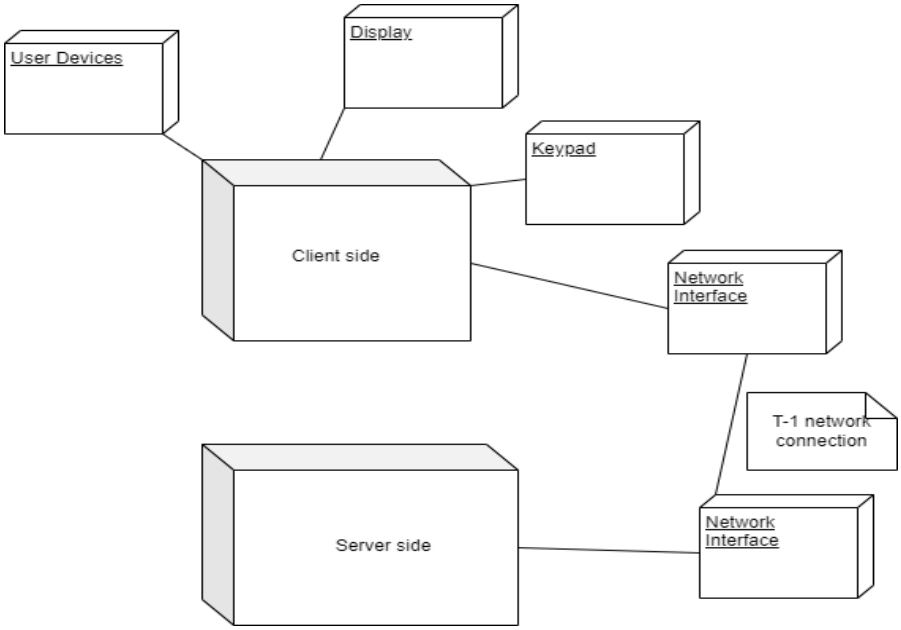

Figure 3.9 Deployment diagram

- Client-Side Device: Represents any end-user device that can run a bare minimum web browser capable of handling video streaming. This could be a desktop, laptop, tablet, or mobile device. There are no specific hardware requirements as long as the device can support the browser functionalities.
- Network Interface: Represents the network interface on the client-side device, which connects to the internet.
- Network: Represents the internet connection that connects the client-side device to the application server.

- Application Server: Hosted on Vercel's free tier, the server has 8192 MB of memory, 23 GB disk space, and a 2 GHz CPU. It handles user requests, processes business logic, and interacts with the database.
- Internal Network: Represents the network infrastructure within the deployment environment that connects the application server and the database.
- Database Server: Stores data related to users, lessons, quizzes, progress, and other relevant information. It can be a dedicated server or a managed database service.

It's important to note that the hardware specifications mentioned (2GHz processor and 8192 MB memory) are limit on vercel’s free tier and suffice for this project. While this deployment model can work with such specifications, it may result in reduced performance, especially if there are concurrent users or resource-intensive operations. Consider upgrading the hardware for better performance if possible. Additionally, this deployment model assumes a simple client-server architecture, but in a real-world scenario, there may be additional components such as load balancers, firewalls, and caching mechanisms to enhance performance, security, and scalability.

CHAPTER 4

ANALYSIS AND RESULT

4.1 System Testing

System testing for the Learniverse application involves a comprehensive evaluation of its functionality and performance to ensure it meets the specified requirements and provides a seamless user experience for both students and teachers. The testing will focus on two primary types: functional testing and performance testing.

- Functional Testing

  Functional testing verifies that each function of the software operates in conformance with the required specification. For Learniverse, this include:

  - Course Publication: Ensure that teachers can successfully create and publish courses, including the ability to add chapters and detailed descriptions.
  - Student Enrollment: Test the enrollment process for students to ensure they can register for courses without issues.
  - Content Accessibility: Verify that students can access course materials and chapter descriptions as intended.
  - MCQ Generation and Functionality: Assess the AI-generated multiple-choice questions (MCQs) to ensure they are relevant to the chapter content and function correctly (e.g., displaying questions, recording answers).
  - Feedback Mechanism: Check that students receive immediate feedback on their MCQ performance, including correct answers and explanations for incorrect responses.
  - User Interface Testing: Validate that the user interface is intuitive and user-friendly for both students and teachers.
  - Error Handling: Test how the system handles invalid inputs or errors during course publication, enrollment, and assessments.

- Performance Testing

  Performance testing evaluates the application's responsiveness, speed, scalability, and stability under various conditions. For Learniverse, the following aspects are tested:

  - Load Testing: Simulate multiple users accessing the platform simultaneously to ensure it can handle a high volume of students enrolling in courses and taking assessments.
  - Stress Testing: Determine the application's breaking point by exceeding normal operational capacity to identify how it behaves under extreme conditions.
  - Response Time: Measure the time taken for various actions, such as loading course content, submitting answers, and receiving feedback, to ensure they meet acceptable thresholds.
  - Scalability Testing: Assess how well the system scales with an increasing number of users and data volume, ensuring that performance remains stable.
  - Reliability and Recovery: Test the system's ability to recover from failures or crashes and verify that data integrity is maintained.

4.2 Outcome

The outcome of this project has been the creation of a strong AI-powered educational platform designed for Nepal Engineering Council license examination candidates. Through incorporating AI to generate specific MCQs on each chapter, the initiative has yielded an individualized learning method, catering for the needs of diverse learners. By targeting areas that need improvement in addition to making their overall preparation more effective and efficient, this creative approach plays a crucial role. The project has established a user-friendly tool which is at the same time influential in making education technology in Nepal better.

4.2 Evaluation

This project has been evaluated as having a marked positive effect on the learning among contestants for Nepal Engineering Council license examination. The AI MCQs, based on continuous testing and user feedback, were observed to drive learning reinforcement in the best possible way as it helped learners remember important concepts better. Quizzes were

structured in a way so as to provide instant feedback and allowed the student to focus on questions from the topics needed improvement which saved time during study hours.

It was concluded that the user interface of the system is very simple and even it does not need any technical knowledge to understand. One of the most crucial innovations in this regard was that they integrated an AI to generate questions from chapter description, which ensured that whatever content they create is directly related to what you would need for your exam syllabus.

A few areas of improvement were highlighted, though. The AI proved to be very good at generating new questions but there were cases when it could do a better job of matching the question difficulty with how far ahead or behind on the learning curve a student was. Moreover, extending the platform to more interactive solution types like video demos or peer discussions would add another dimension of learning experience.

To conclude, the project has been considered successful in achieving its goals to deliver personalized learning with an AI-based tool. These learnings will be used to feed back into subsequent versions of the platform and allow us to further refine how best we can support learners.

CHAPTER 5

CONCLUSION AND FURTHER DISCUSSION

5.1 Conclusion

Learniverse is a transformative platform designed to prepare students for the Nepal Engineering Council license exam, particularly for those in software and computer engineering. By incorporating advanced AI to generate tailored Multiple Choice Questions, it fills critical gaps in traditional study materials and offers a focused, personalized approach to exam preparation. This innovation enhances content organization and progress assessment, reducing the stress associated with conventional tools. The platform underwent extensive testing, including user feedback sessions and performance evaluations, to ensure its reliability, adaptability, and effectiveness. Ultimately, Learniverse equips students with the skills needed to excel in their exams and future careers.

5.2 Limitation and Future Recommendation

Learniverse is more than just a tool; it is a transformative approach to exam preparation that addresses a critical need for students pursuing the Nepal Engineering Council License Exam, especially those in software and computer engineering.

5.2.1 Limitation

While Learniverse represents a significant advancement in personalized exam preparation, there are certain limitations to the platform that must be acknowledged:

- Limited Subject Scope: Currently, Learniverse is focused on the software and computer engineering fields. While this specialization allows for deep customization, it limits the platform's applicability to students in other engineering disciplines or fields of study.
- Resource Availability: The quality of personalized questions depends on the availability of comprehensive and up-to-date course materials. If course content is not

readily available or if it varies significantly across different institutions, the platform may struggle to deliver consistent and relevant questions.

5.2.2 Future Recommendation

To enhance the effectiveness and reach of Learniverse, several future recommendations can be considered:

- Expansion of Subject Areas: To broaden its impact, Learniverse could expand its offerings to include other engineering disciplines or even other fields of study. This would make the platform accessible to a wider audience and allow for cross-disciplinary learning.
- Integration with Educational Institutions: Collaborating with educational institutions could help standardize the content used by Learniverse, ensuring that the AI generates questions that are consistent with the curriculum. Partnerships with universities and colleges could also facilitate the continuous updating of course materials.

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APPENDICES

- User Guidance
    - This is the main Homepage where users after successful login will be redirected. Here is one carousel for the motivational quote and two cards that shows the count of the in progress courses and the completed courses and the cards of courses that users have enrolled into.

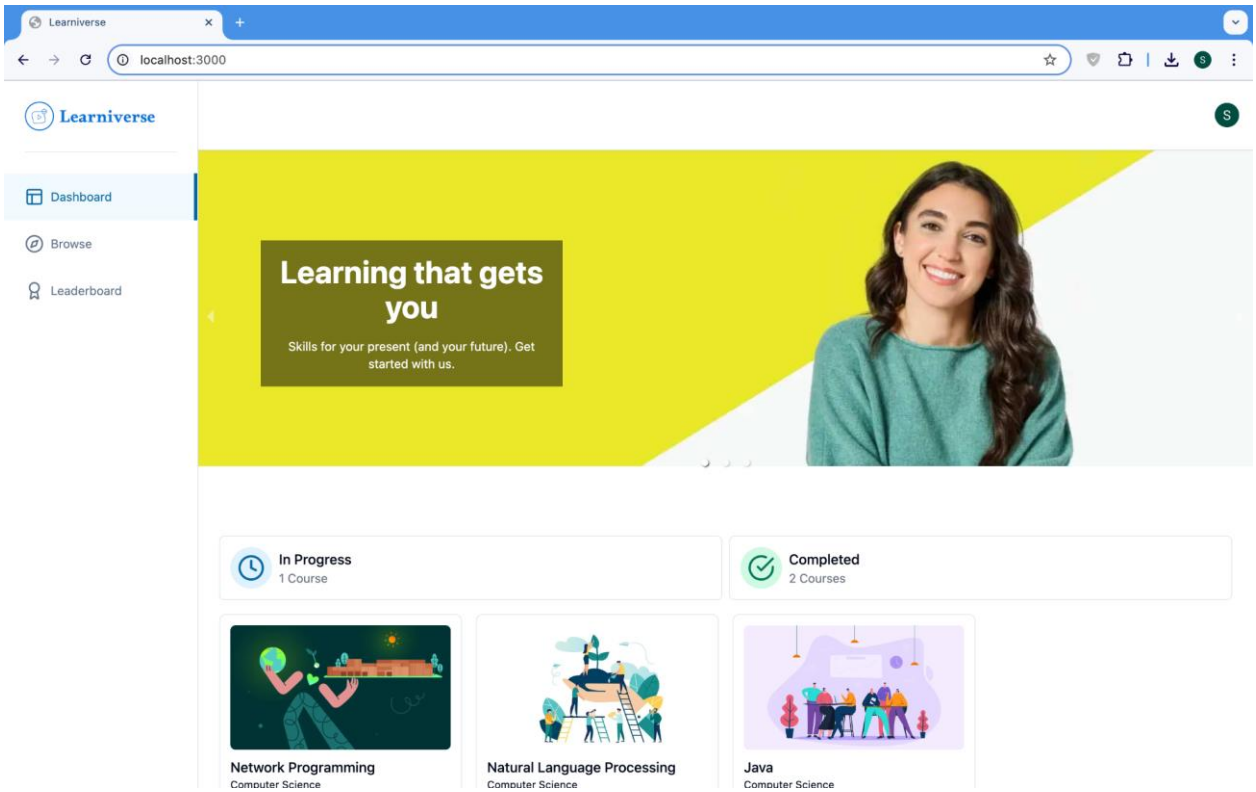

Figure a Home page

- This is the Browse section where user can browse the courses that has been published in this platform

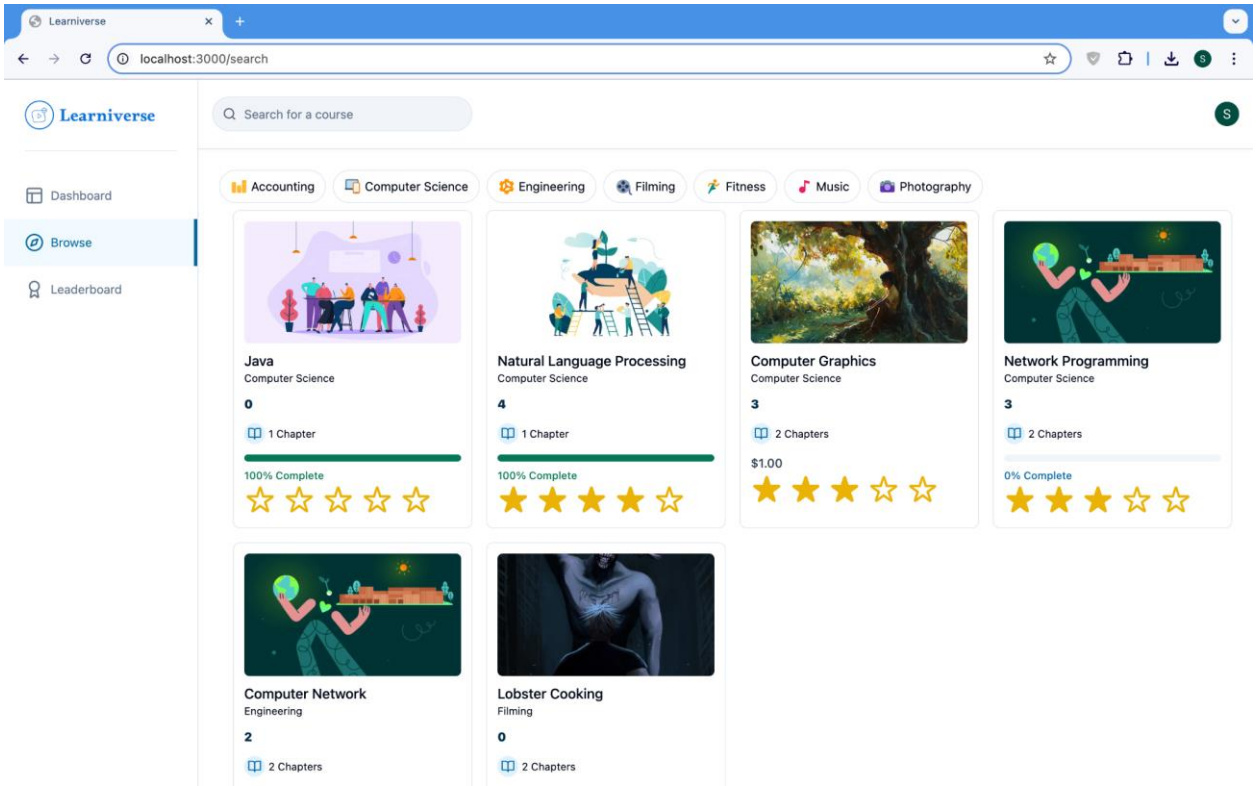

Figure b Browse Page

- When users select the course then they will be redirected to this page where they can see the chapters. If there is a free chapter for preview they can watch video without enrollment but if chapter is not free then user have to enroll the course by doing the payment for the access

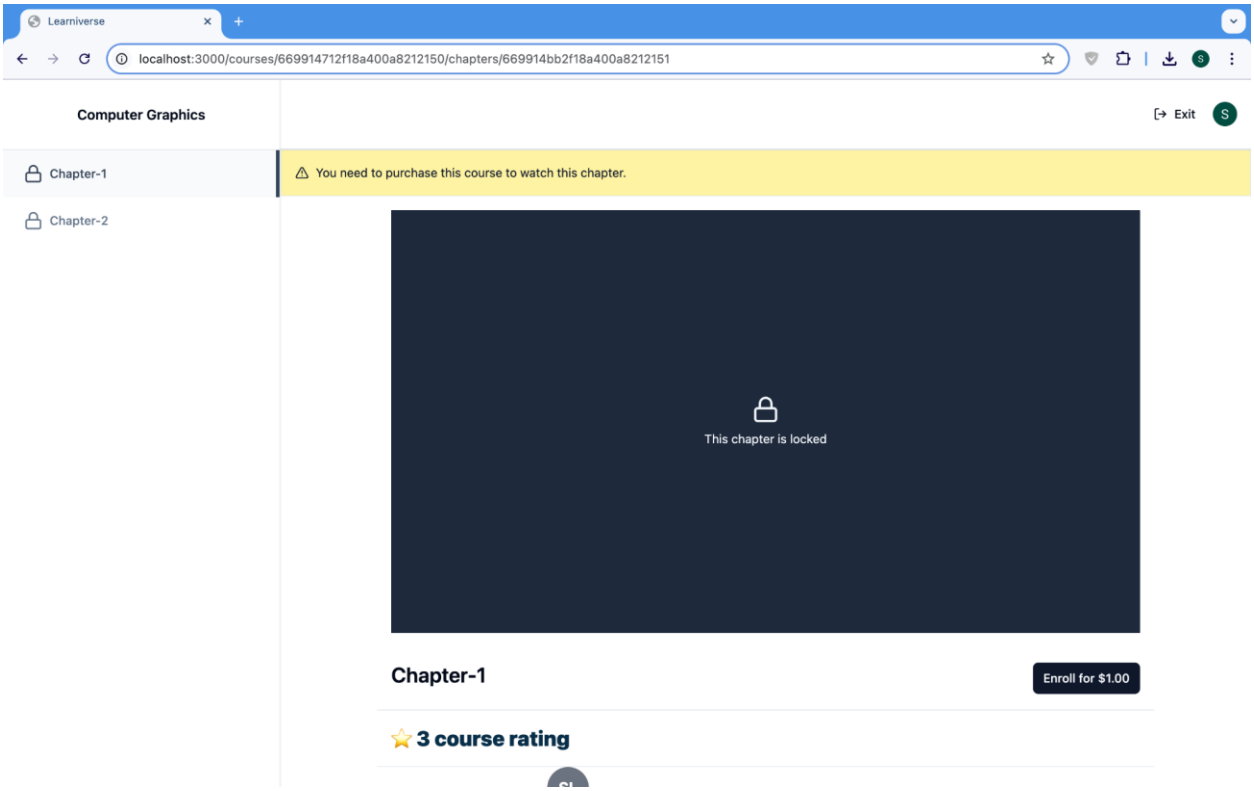

Figure c Course page

- For enrollment users have to click on the enroll for button then it will redirect to this payment portal, here we have used Stripe for the payment process. By filling the credit card's details, users can purchase the course forever.

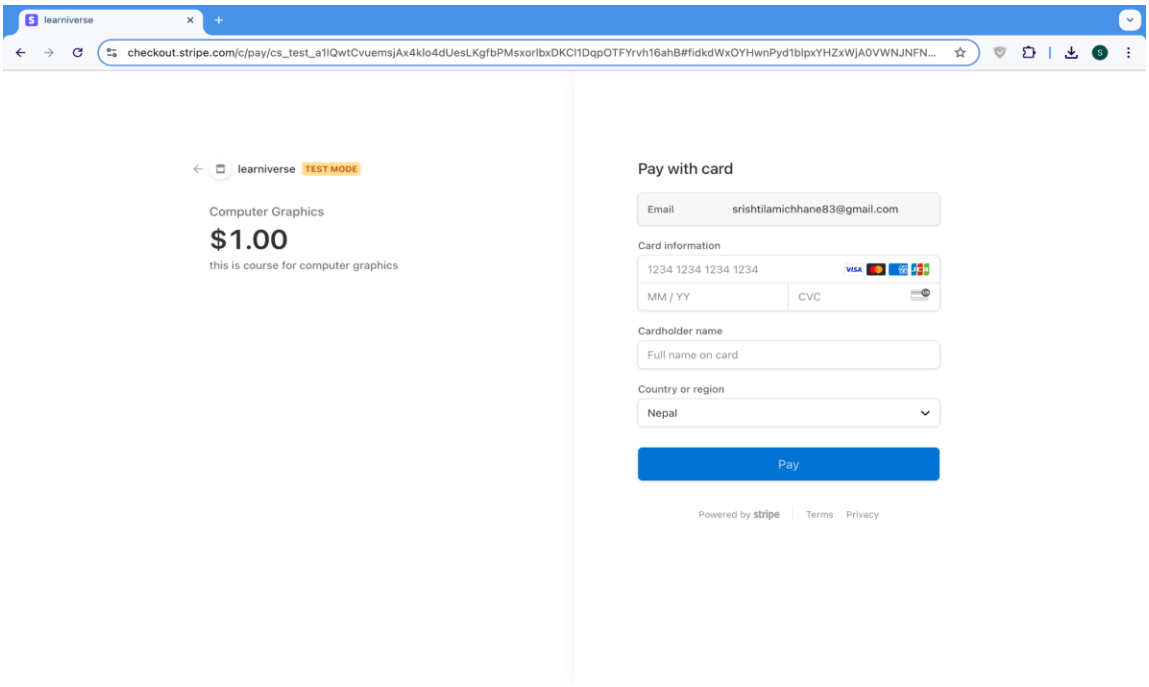

Figure d Payment Portal

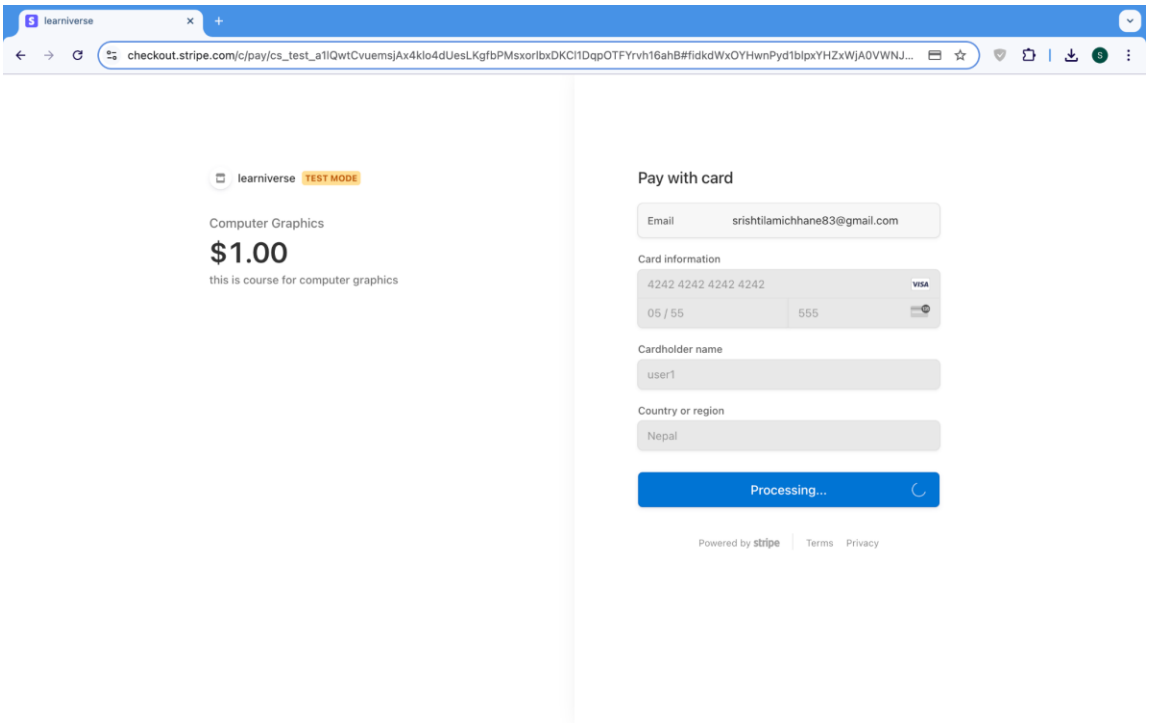

Figure e Payment Processing

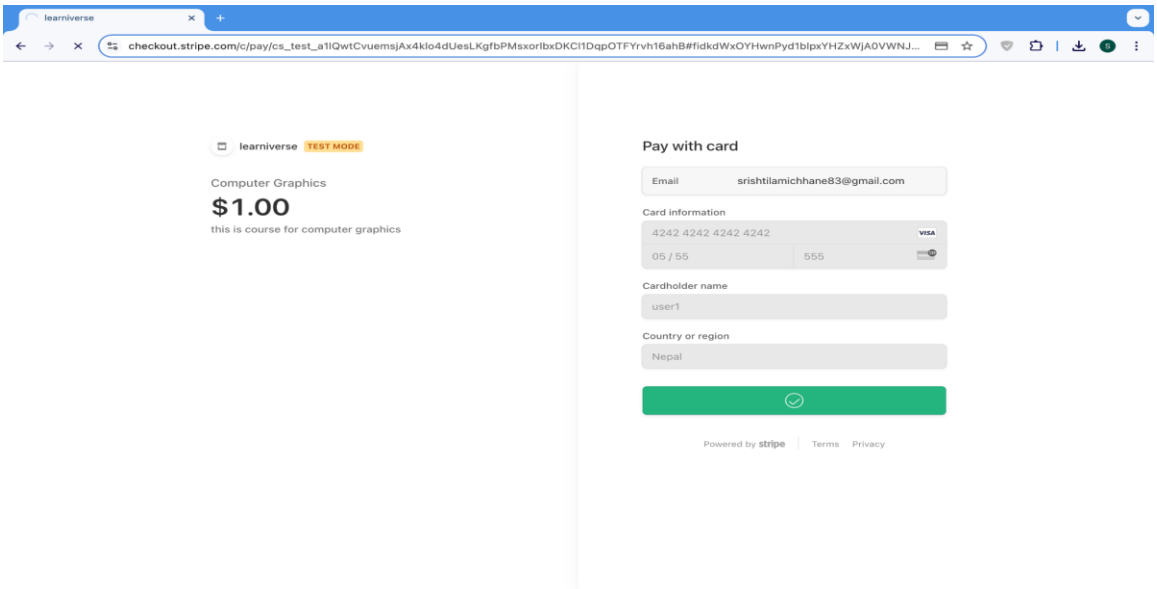

Figure f Successful Payment

- After the successful payment user will have access to the course description, video will be unlocked and the attachment as well.
- And after the successful completion of the course users will be able to see the confetti , this feature is added to praise them for studying.

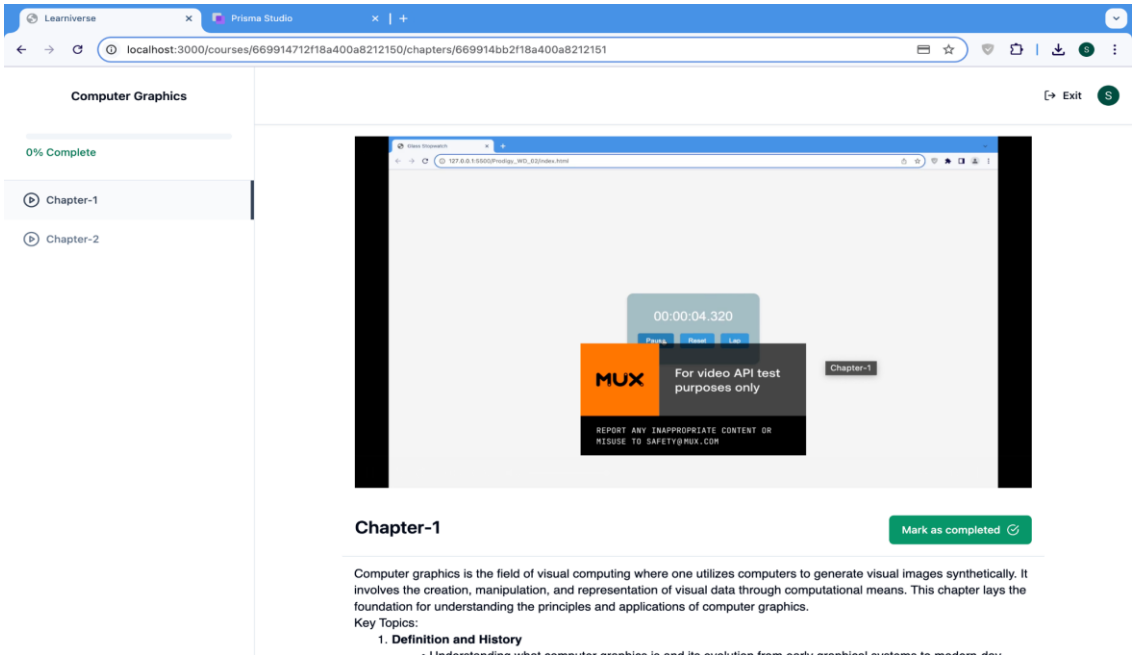

Figure g Chapter Page

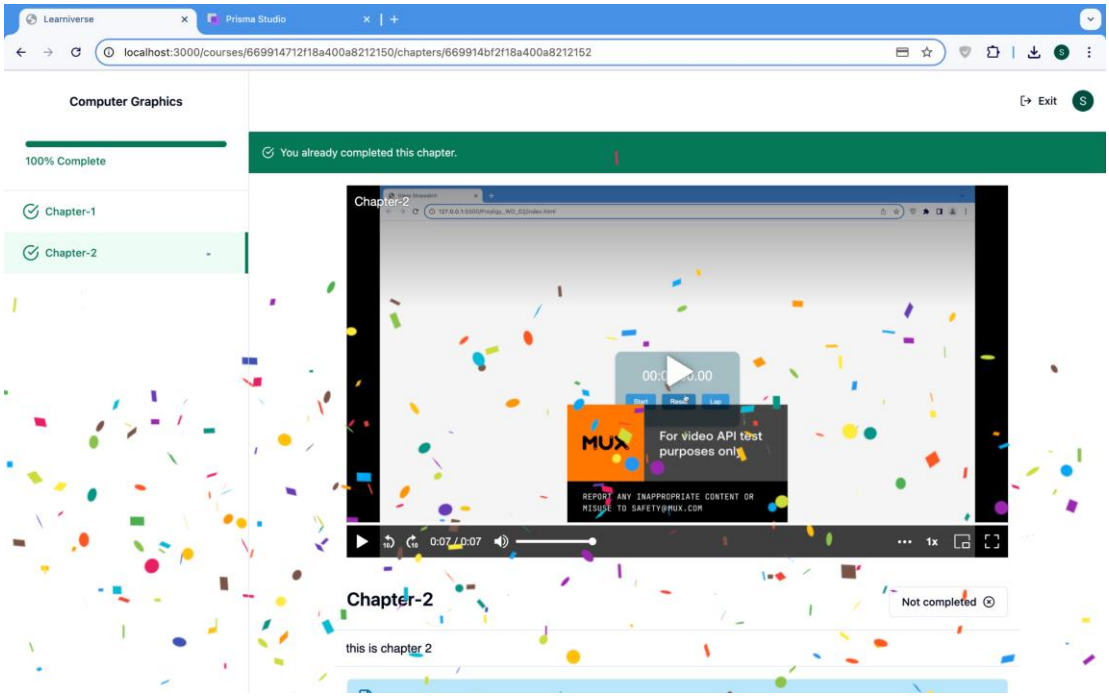

Figure h Chapter Completion

- Once the student completes the courses provided resources by studying then there is a generate mcq button right below the description. This is the main feature of this application. This will generate the mcqs related to the chapter's descriptions. There are three category initially user will be able to take beginner level's mcqs test and after passing that they will be able to take intermediate level mcqs test and after passing this level they will be able to take advance level finally, if they don't gain the sufficient marks in each level they they have to retake the same level exam, they won't be able to go to another level without passing the current level.

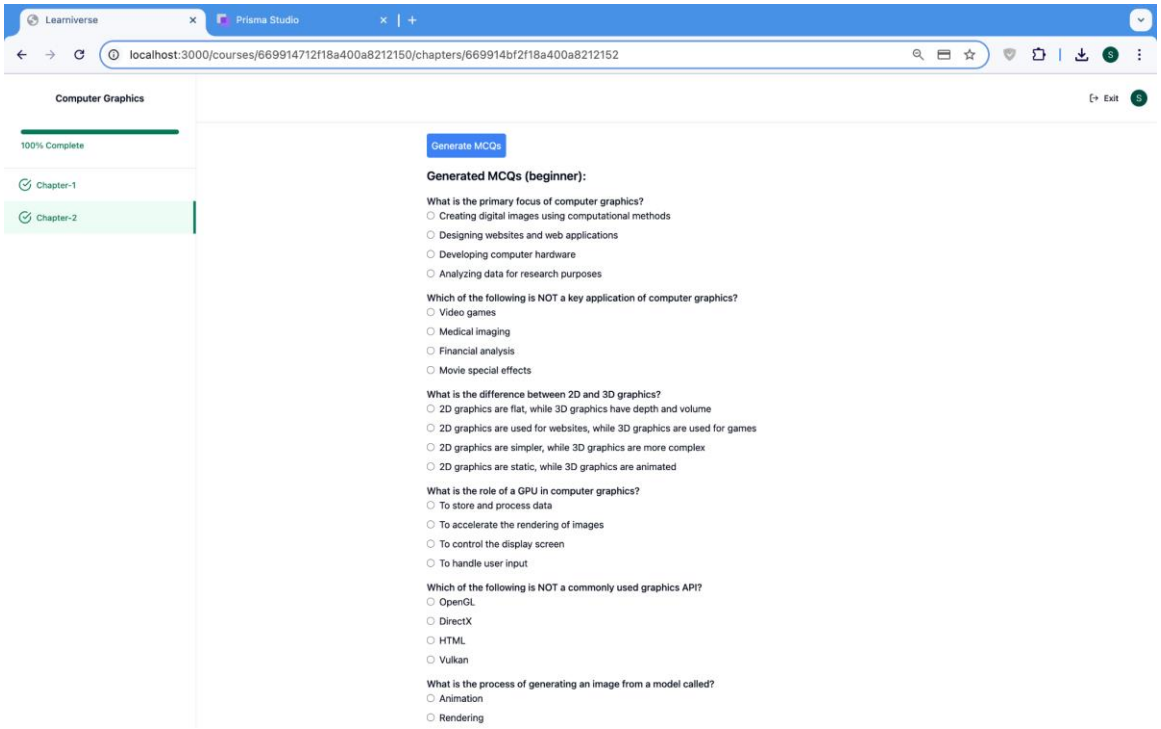

Figure i MCQ for Beginner

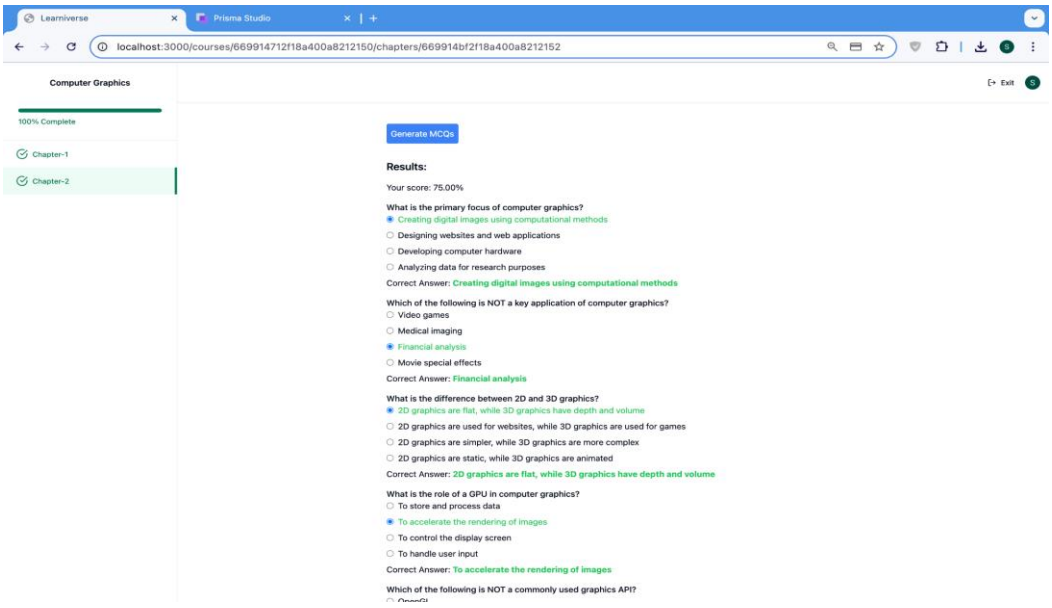

Figure j Beginner's Result

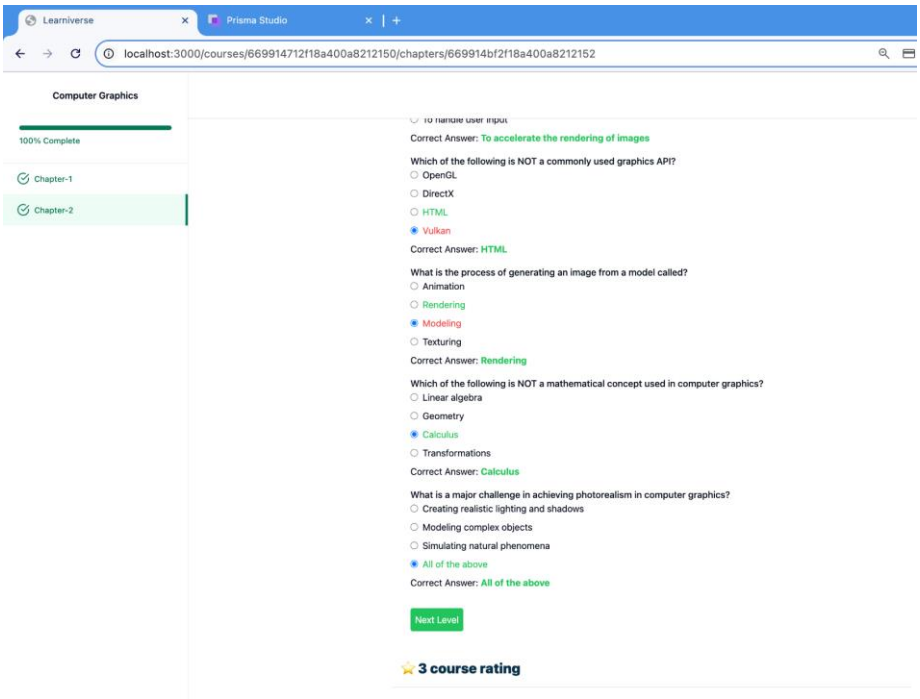

Figure k Next Level

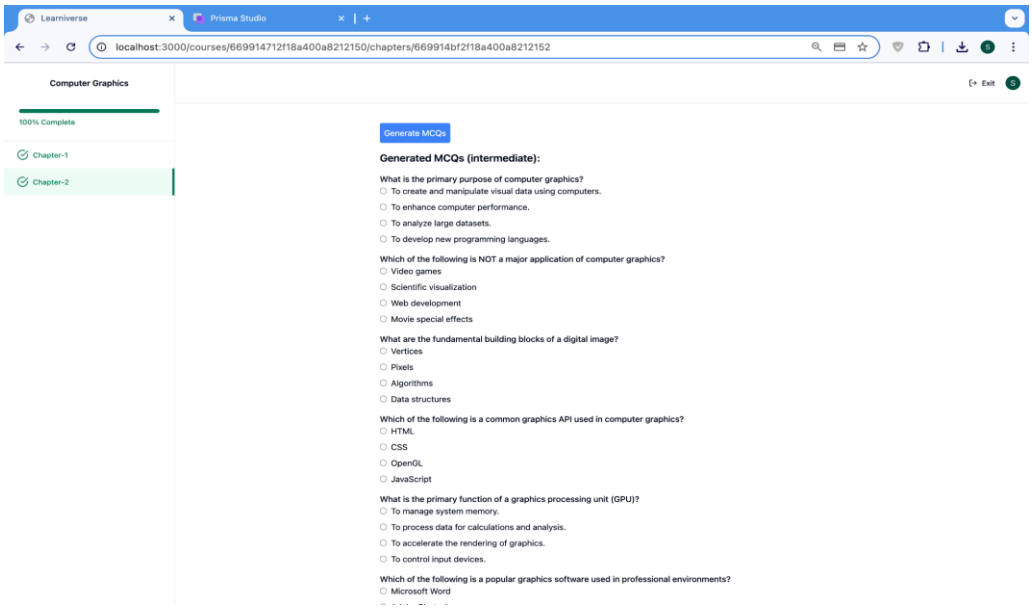

Figure l MCQ for Intermediate

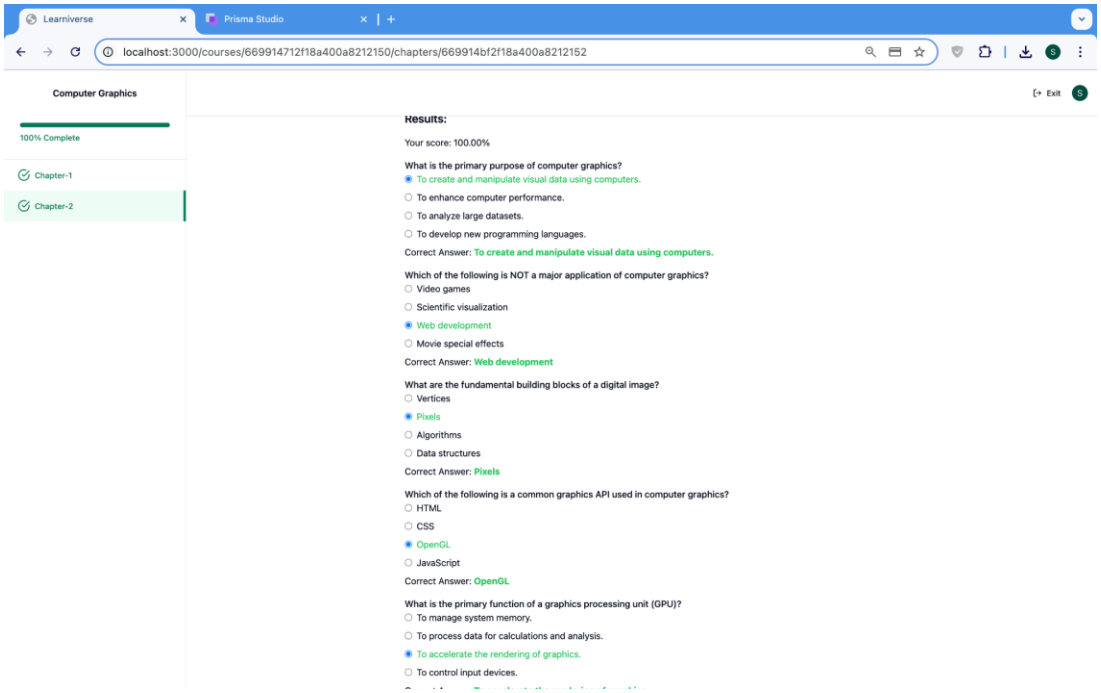

Figure m Intermediate Result

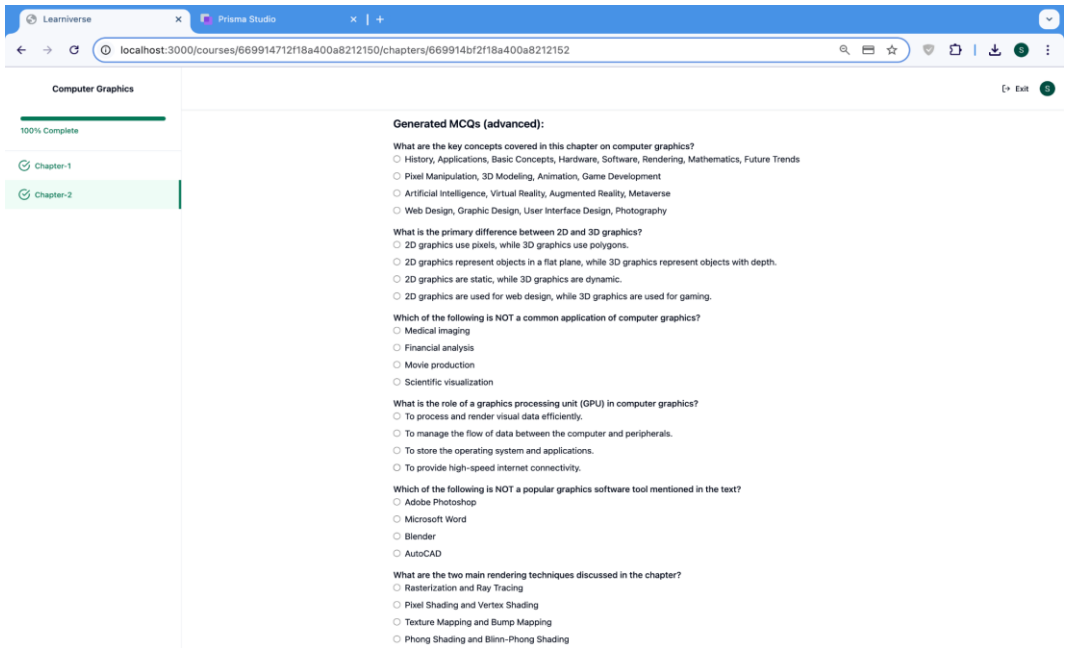

Figure n MCQ for Advance

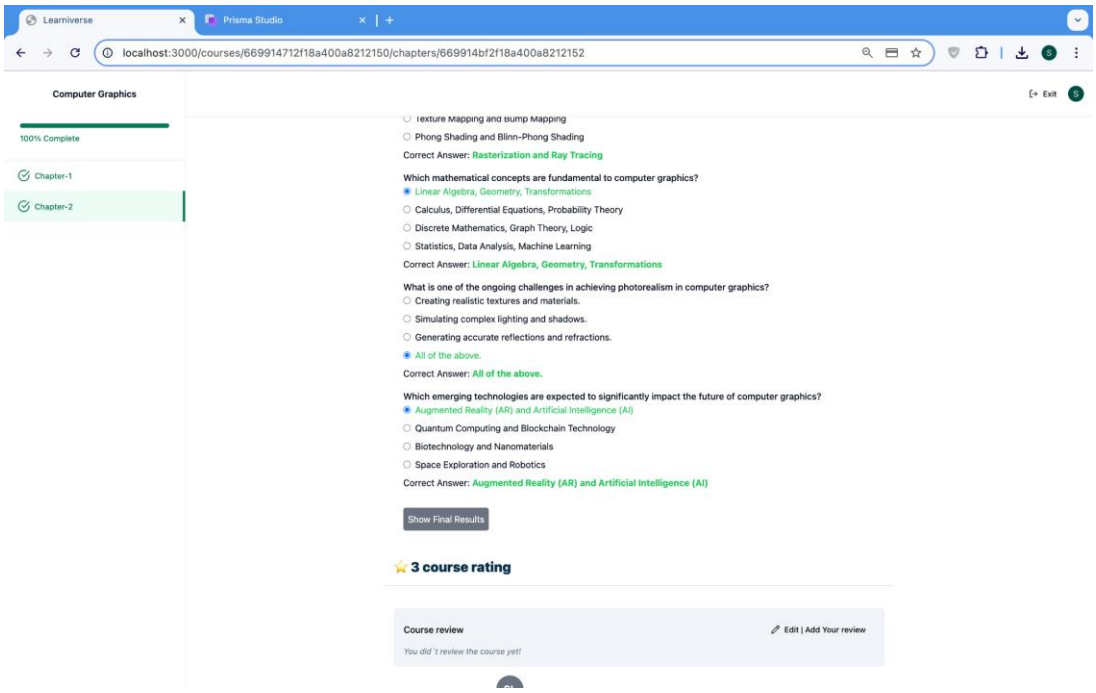

Figure o Advance MCQ finished

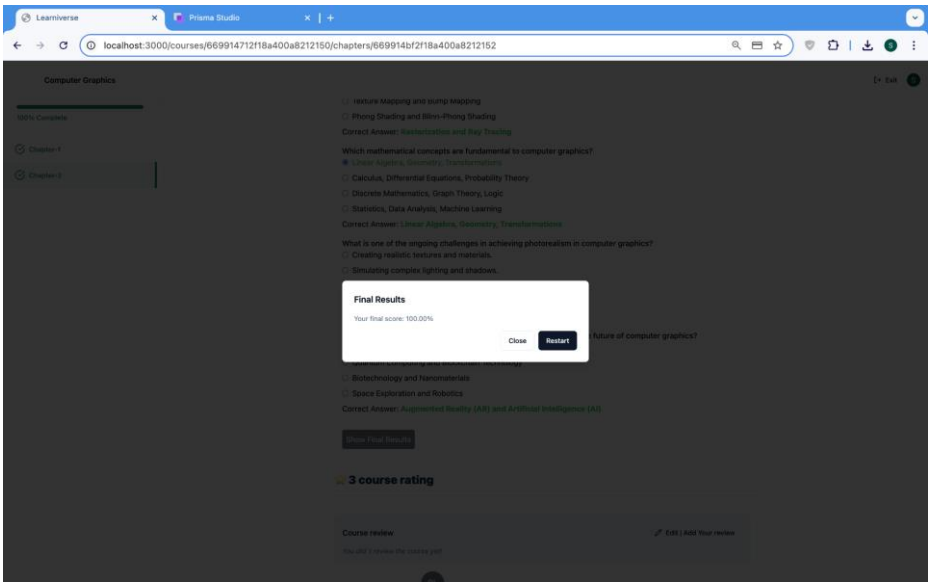

Figure p Final Result

- Users can give the ratings to the course and write reviews for the course too.

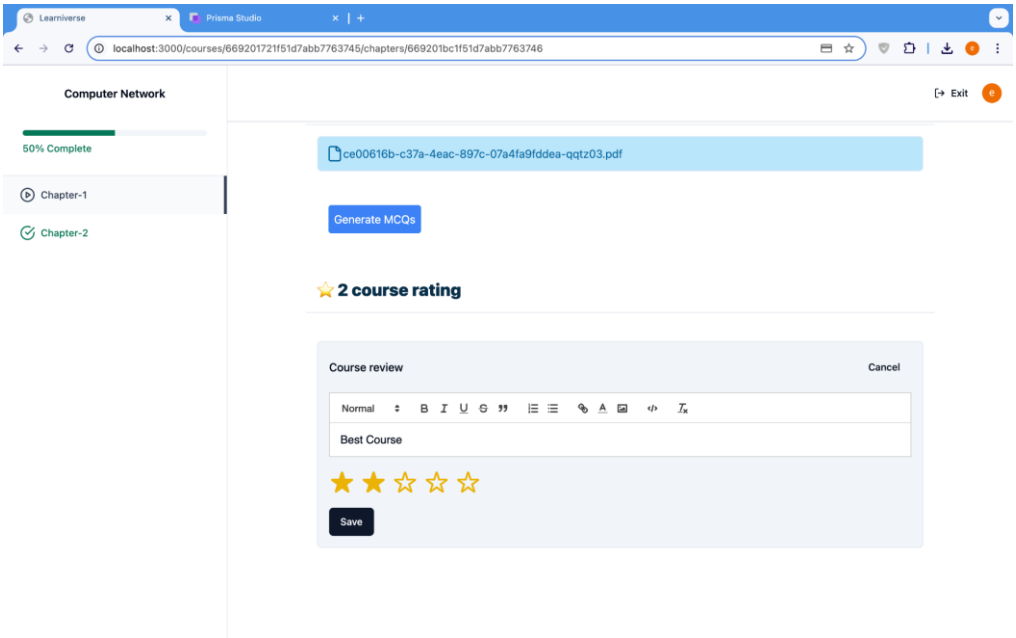

Figure q Rating and Review Section

- There is a leaderboard where users can see their progress. This is added for the encouragement to the student as a gamification feature.

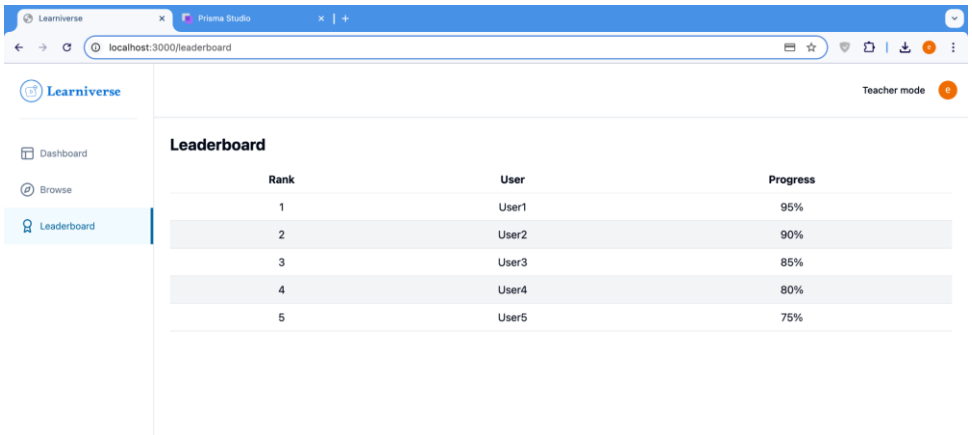

Figure r Leaderboard

- Teacher's Guidance
    - Only specific ids can be logged in as teachers, so the teacher id can be able to see the teacher mode from the dashboard and on clicking on that button they will be redirected to the teacher's courses section. Here is the list of courses that the teacher has published in the form of a table.

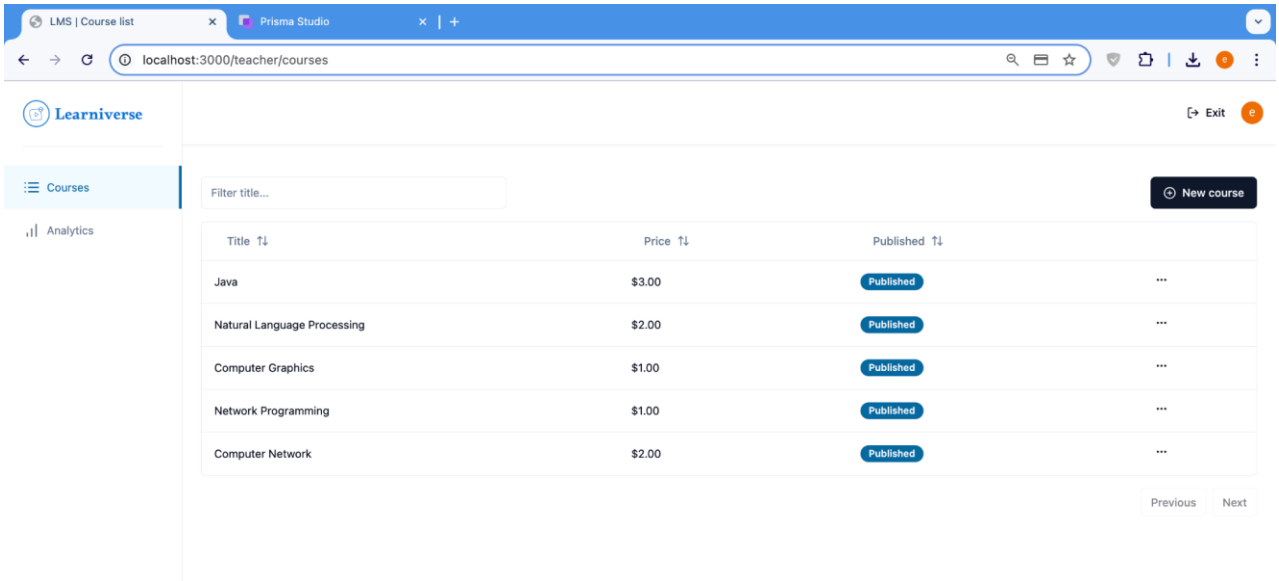

Figure s Table of published courses

- By clicking on New Course button teacher can create the new course by giving the course title

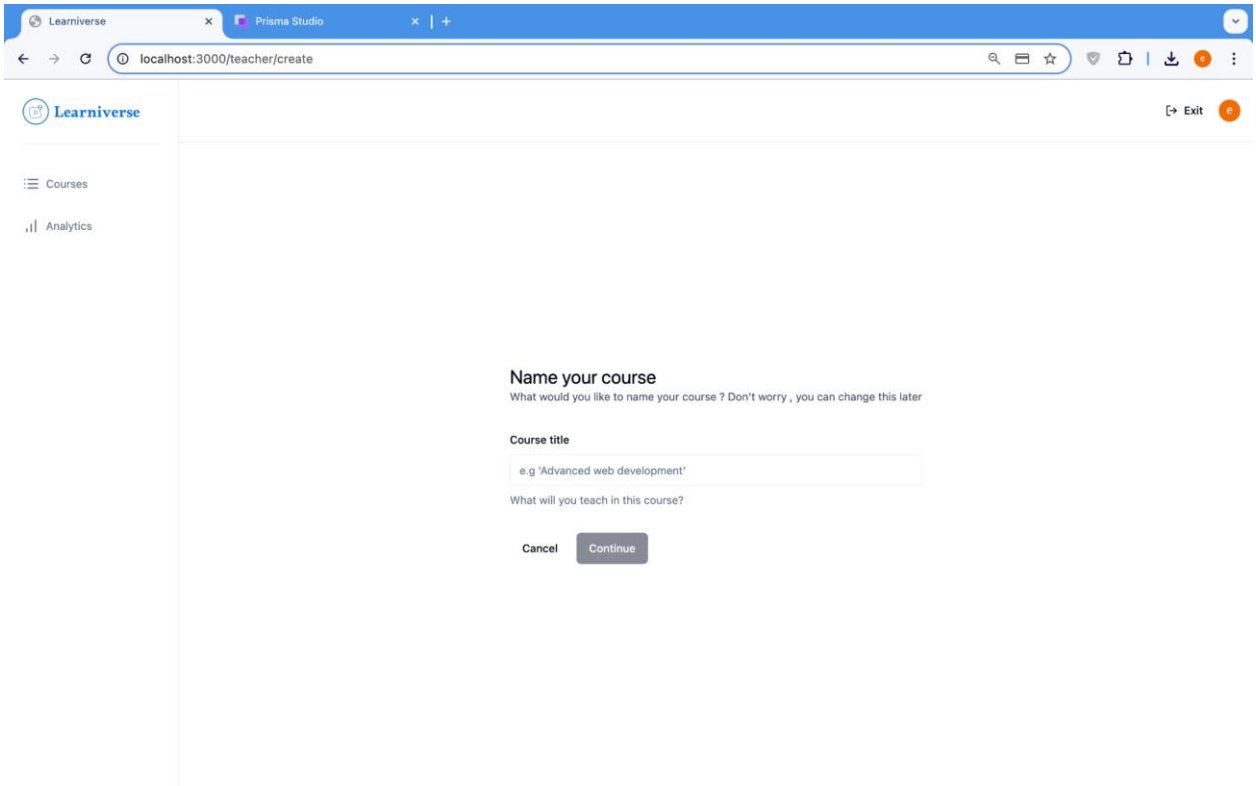

Figure t Create Course

- When a course is created then the teacher will be redirected to the courses pages where they need to fill up the descriptions of the courses given.

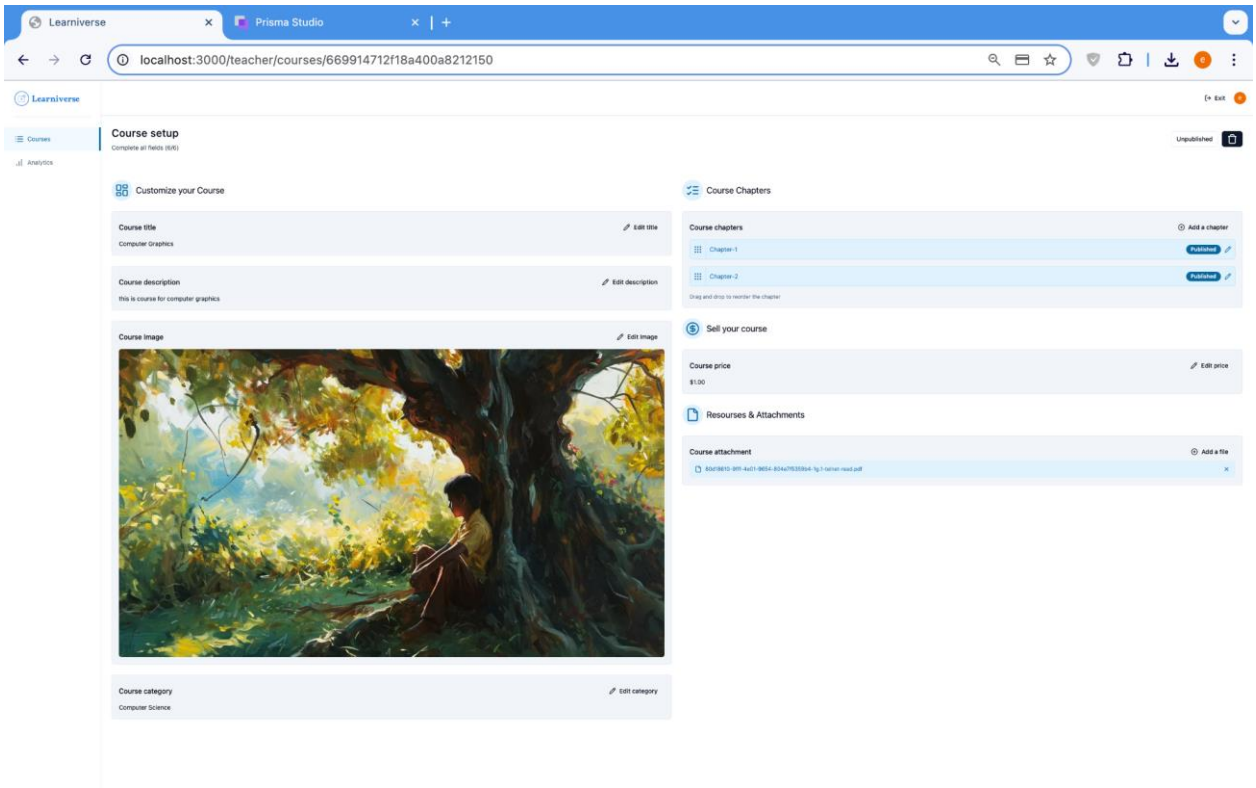

Figure u Course Fields

- Courses include chapters. By clicking the add chapter teacher can add chapter. When the teacher adds some chapters they are draggable and can be sorted in any manner as they want. After clicking on the pen icon of the chapter, the teacher will be redirected into the chapter's pages, where they have to fill up all the necessary resources that are needed for this chapter to be published.

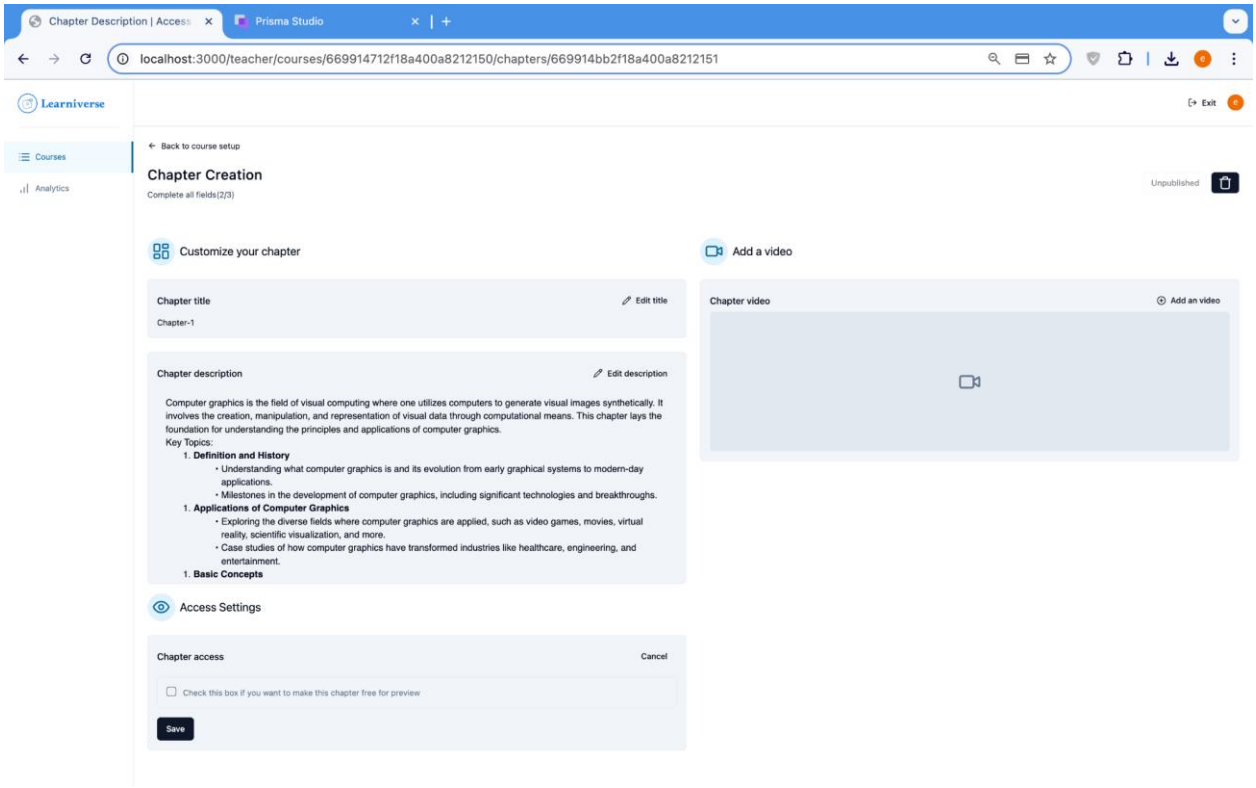

Figure v Chapter Fields

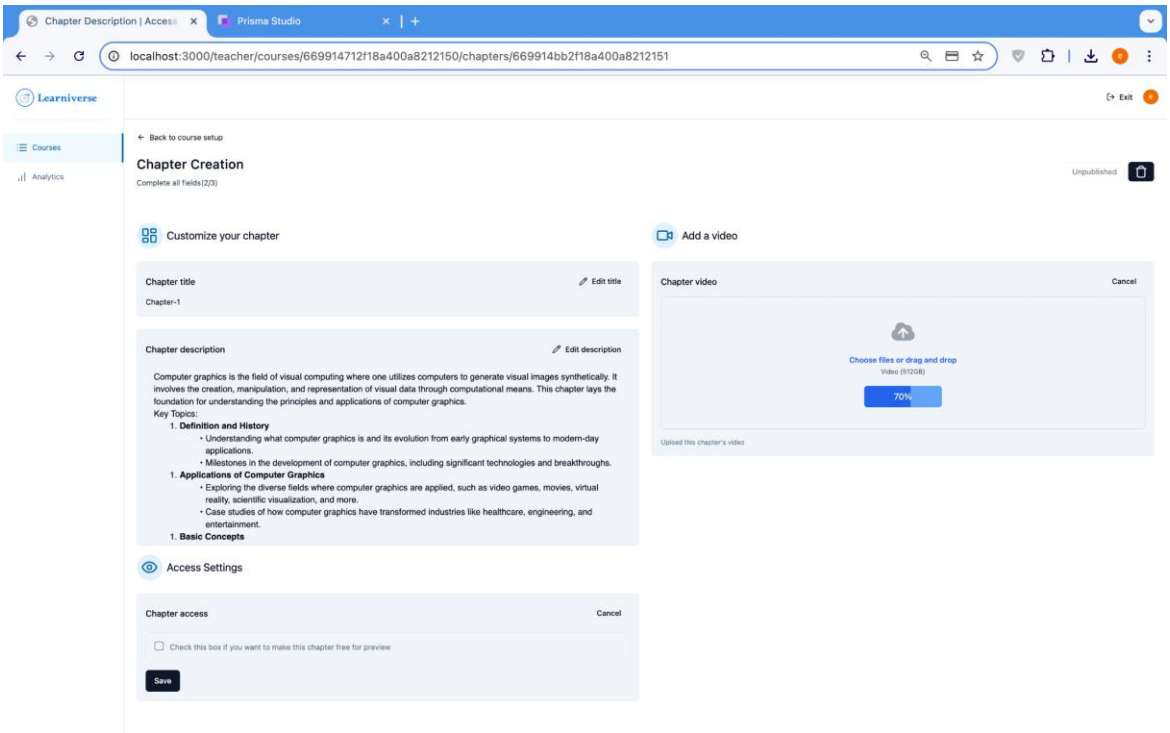

Figure w Uploading Video of Chapter

- The unpublished course and the chapters have that warning banner that this will not be visible to students, actually without publishing at least one chapter the course cannot be published. To enable the publish button all the six out of six fields should be filled.

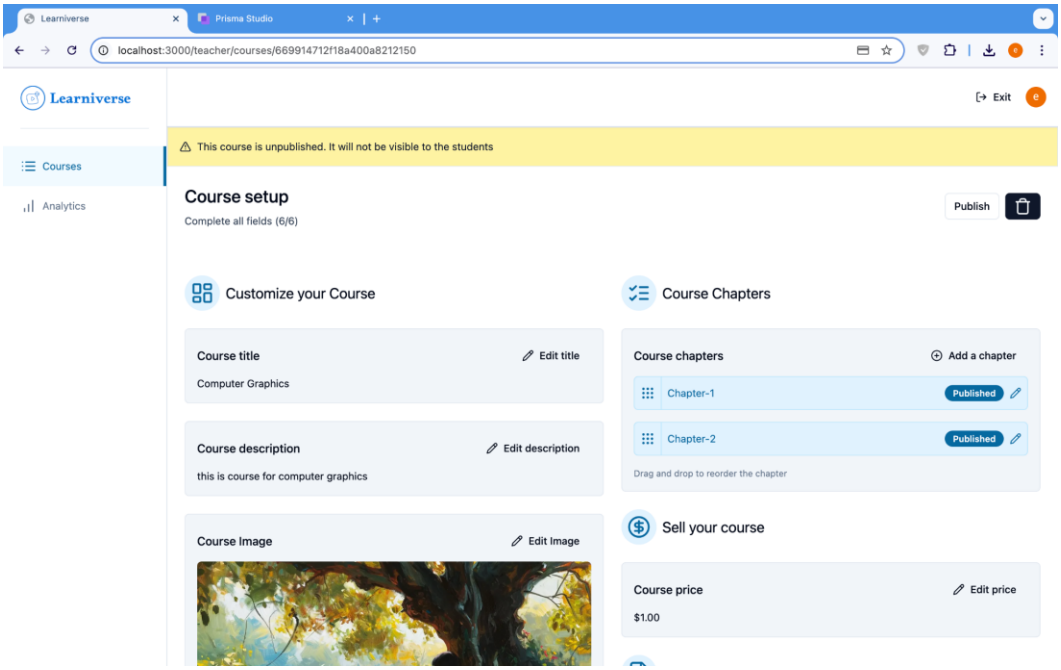

Figure x Unpublished Course

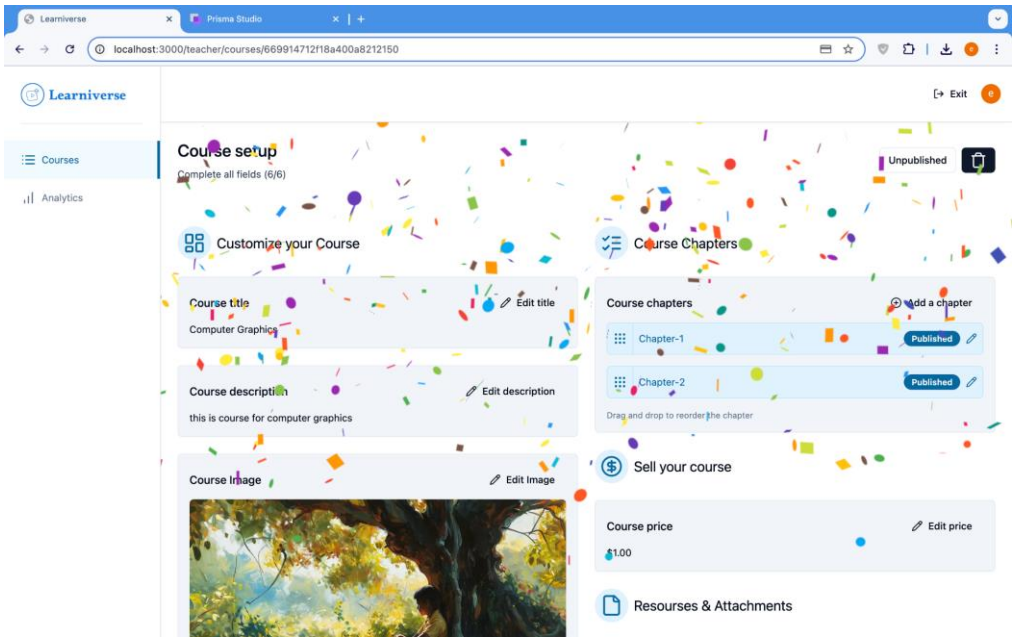

Figure y Published Course

- Teacher will be able to see the analytics of the course that has been published

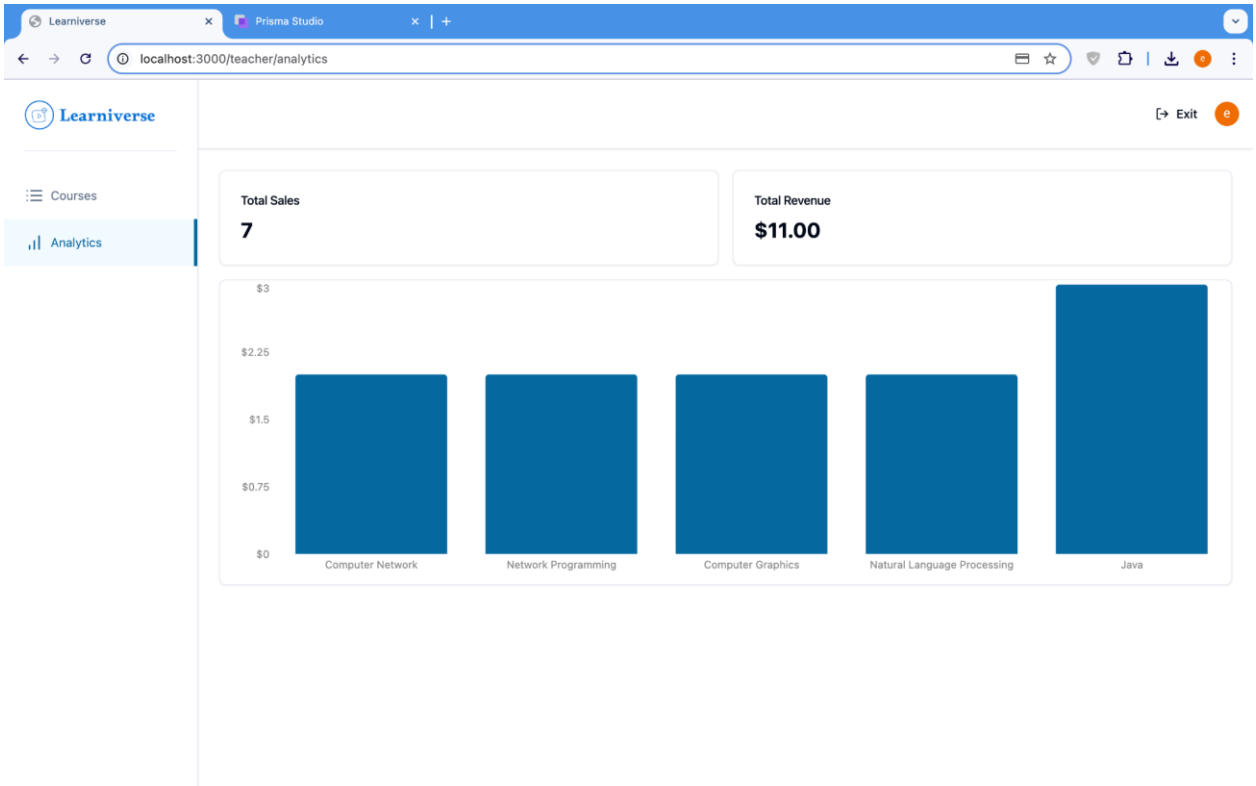

Figure z Analytics Section